Erasmus University Rotterdam

Erasmus School of Economics

Master Thesis Econometrics & Management Sciences

Quantitative Finance

Measuring Attention to Physical Climate Risk: A Text-Based Index Separating Salience from Surprise

Adam Mc Guire (770113)



Supervisor:	Dr. Annika Camehl
Second assessor:	Dr. Mariia Artemova
Date final version:	16th August 2026

The content of this thesis is the sole responsibility of the author and does not reflect the view of the supervisor, second assessor, Erasmus School of Economics or Erasmus University.

Abstract

This paper constructs a daily index of physical climate risk attention for the United States from 2008 to 2025, decomposed into a persistent salience component (the Structural Salience Index) and a stationary surprise component (the Climate Surprise Proxy). Articles from seven national outlets are filtered using the Intergovernmental Panel on Climate Change and the Task Force on Climate-related Financial Disclosures physical risk vocabulary, with transition risk terms excluded at the retrieval stage. A comprehensive validation exercise confirms that the index captures a genuine distinct physical risk information channel with 89.5% precision and is uncorrelated with other text-based indices. In financial markets, the Treasury curve exhibits a robust null response across all maturities and horizons. However, evidence from equity markets indicates that the signal is present but more provisional. Unconditional exposure effects are insignificant, but conditioning on structural salience reveals amplified reactions to climate surprise in two industries, Food and Utilities.

Contents

1 Introduction **1**

2 Measuring Climate Risk **2**
2.1 Why Physical and Transition Risk Need Separation 2
2.2 Existing Climate Risk Measures 3
2.3 Constructing a Credible Text-Based Index 4

3 Index Construction **5**
3.1 Corpus Construction 5
3.2 Article-Level Scoring 6
3.3 Novelty-Weighted Aggregation 7
3.4 Index Decomposition: SSI and CSP 8

4 Properties of the PCRA Index **8**
4.1 The Structural Salience Index 9
4.2 The Climate Surprise Proxy 10
4.3 Salience and Surprise Combined: the PCRA 11

5 Validation of the PCRA Index **12**
5.1 Mechanical Robustness 12
5.2 Layered Construction Validity 13
5.3 Event Validity 15
5.4 Human Coder Validity 16
5.5 Placebo Validity 18
5.6 Discriminant Validity 19
5.7 Principal Component Analysis 21

6 Treasury-Curve Response Evidence **21**
6.1 Baseline Local Projection Specification 22
6.2 Main Result: A Robust Null 22
6.3 Evaluating Alternative Explanations 23
6.4 Specification choice 24
6.5 Sign Counts, Events, Subsamples, and Asymmetry 25

7 Industry-Level Equity Evidence **26**
7.1 Linear Exposure Rankings do not Explain the Cross-Section 26
7.2 Beta-Sort and Industry Level Sensitivities 28
7.3 Salience Conditioned Response 29
7.4 Firm Level Corroboration of the Industry Results 31
7.5 Economic Magnitude 32

8 Conclusion **33**
8.1 Discussion and Extensions 34

References **36**

Appendix **40**

List of Abbreviations

Core Index Terms

PCRA	Physical Climate Risk Attention (index)
SSI	Structural Salience Index
CSP	Climate Surprise Proxy
LM-CSP	Loughran–McDonald Climate Surprise Proxy (placebo index)

Comparator Indices

CPU	Climate Policy Uncertainty (index)
EPU	Economic Policy Uncertainty (index)
MOVE	Merrill Lynch Option Volatility Estimate (index)
GPR	Geopolitical Risk (index)

Data Sources

EM-DAT	Emergency Events Database
IPCC	Intergovernmental Panel on Climate Change
TCFD	Task Force on Climate-related Financial Disclosures
AR6	Sixth Assessment Report (IPCC)
WGII	Working Group II (IPCC)

Econometric Methods and Tests

ADF	Augmented Dickey–Fuller (test)
ARCH-LM	Autoregressive Conditional Heteroskedasticity Lagrange Multiplier (test)
BIC	Bayesian Information Criterion
CUSUM	Cumulative Sum (test)
EWMA	Exponentially Weighted Moving Average
HAC	Heteroskedasticity and Autocorrelation Consistent (standard errors)
KPSS	Kwiatkowski–Phillips–Schmidt–Shin (test)
OLS	Ordinary Least Squares
PCA	Principal Component Analysis
SUR	Seemingly Unrelated Regression
TF-IDF	Term Frequency–Inverse Document Frequency

Fama–French Factor Models

FF3	Fama–French Three-Factor (model)
FF5	Fama–French Five-Factor (model)
FFC	Fama–French–Carhart Four-Factor (model)
FF12	Fama–French 12-Industry (classification)
FF49	Fama–French 49-Industry (classification)
UMD	Up-Minus-Down (momentum factor)

1 Introduction

The economic consequences of climate change materialise through two distinct channels: realised physical damage and changes in expectations about future risk (IPCC, 2021). While the damage effects from floods, wildfires, and extreme temperatures are increasingly documented (Dell et al., 2012), the expectations these events generate remain less directly observable. The International Monetary Fund and the Network for Greening the Financial System recognise that climate related financial risks are insufficiently incorporated into asset valuations, leaving physical climate risk systematically underpriced (International Monetary Fund, 2020; Anyfantaki et al., 2025). Evidence from multiple asset classes reiterates that equity markets respond imperfectly to physical risk exposure (Faccini et al., 2023), sovereign spreads are linked to environmental risks (Capelle-Blancard et al., 2019), and long horizon bond returns may become increasingly sensitive to physical damage (Lorans and Moussavi, 2021).

Understanding how markets respond to physical climate risk necessitates a measure that isolates changes in information specific to that risk. Although existing text-based climate indices have become widely used for this, two limitations complicate their interpretation in the context of physical climate risk. First, most indices conflate physical and transition climate risks by combining coverage of extreme weather events with discussions of regulatory policy, technological change, and decarbonisation efforts (Gavriilidis, 2021; Jung and Hannaoui, 2024). Second, these indices treat repeated reporting as equivalent to the arrival of new information. As a result, such indices cannot be interpreted as measuring either channel distinctly, since transition related signals tend to dominate aggregate indices (Faccini et al., 2023).

This thesis addresses both limitations through the construction of a high-frequency measure of physical climate risk, the Physical Climate Risk Attention (PCRA) index. The index is built from US newspaper coverage for the period 2008–2025, as the United States combines substantial exposure to physical hazards with deep sovereign and equity markets. Articles are restricted to physical risk vocabulary derived from the Intergovernmental Panel on Climate Change (IPCC) and the Task Force on Climate related Financial Disclosures (TCFD) taxonomies, and transition risk language is excluded at the retrieval stage. At the article level, a relevance based scoring function captures substantive physical hazard content. This content is then aggregated using novelty weights derived from sentence embeddings, discounting semantically repeated coverage.

The resulting PCRA is decomposed into two components corresponding to distinct features of the physical climate risk information environment. The persistent reporting component is captured by the Structural Salience Index (SSI), an exponentially weighted moving average that captures

the persistent baseline of physical climate risk attention. The SSI identifies the slow-moving shifts in attention to climate events that could enter the market's long-run discount rate. The unexpected component is captured by the Climate Surprise Proxy (CSP), which isolates the unanticipated changes in coverage that represent new information entering financial markets. Using an autoregressive model, AR(p), selected by the Bayesian Information Criterion (BIC), the CSP removes predictable attention dynamics to identify short run innovations.

A cumulative six test validation supports the index. The CSP component of the PCRA recalls physical climate hazard events more reliably than raw climate article volume, achieving a 33.8% hit rate against a permutation based null of 27.5%. The index demonstrates high precision, correctly identifying articles about physical climate with 89.5% precision. Principal component analysis further indicates that the PCRA represents a distinct factor, separate from transition risk and policy uncertainty, with absolute correlations with these indices below 0.11 for both. These findings support PCRA as capturing a unique physical climate risk channel, rather than repackaging existing coverage or uncertainty.

Following validation, the PCRA is applied in two empirical settings to test whether physical climate risk information influences asset prices. The results indicate a selective, state dependent channel rather than a uniform pricing effect. Physical climate risk information does not affect US Treasuries, but produces systematic responses in equities when physical exposure and climate salience are jointly considered. High exposure industries, particularly Food and Utilities, exhibit significantly amplified reactions to surprise during high salience regimes ($p = 0.037$ and $p = 0.011$), with consistent directional responses across constituent firms. The effect is economically meaningful but narrow: no identifiable reaction around disaster events and it does not translate to exploitable action strategies. By distinguishing persistent attention from surprise, PCRA establishes a basis for examining when physical climate risk information attains economic relevance.

The thesis proceeds as follows. Section 2 reviews the literature on climate risk measurement and text-based uncertainty indices. Section 3 describes the data and construction of the PCRA index. Section 4 and Section 5 present properties and validation of the constructed PCRA index respectively. Sections 6 and 7 examine sovereign and equity markets. Section 8 concludes.

2 Measuring Climate Risk

2.1 Why Physical and Transition Risk Need Separation

Physical and transition climate risks are economically distinct phenomena operating through different channels, frequencies, and implications for investors and policymakers. Transition risk

is mediated by political and regulatory processes that investors continuously monitor (Krueger et al., 2020), information tends to accumulate gradually through expectations about future regulation and policy intervention. When materialised, the risks can propagate as a negative supply surprise, decreasing output while raising commodity and consumer prices (Gavriilidis et al., 2026).

Physical risk operates through a different mechanism. Extreme weather events cause direct and discrete economic damage, compress regional activity, increase emergency public expenditure, damage infrastructure, and disrupt productive capacity (Zenios, 2023; Cevik and Jalles, 2020). These two categories of climate risk can move in opposite directions. Treating climate risk as a single concept risks obscuring the underlying mechanisms, timing and economic effects.

2.2 Existing Climate Risk Measures

Text-based measurement literature demonstrates that newspaper coverage reliably proxies for unobservable economic uncertainty when vocabulary is chosen with discipline (Gentzkow et al., 2019; Baker et al., 2016). Newspaper attention provides a real-time measure of the information environment through which expectations are formed and revised. In the climate context, attention captures both realised climate damages and also the evolving beliefs about the future climate related risks, relevant to forward-looking financial decision making (Engle et al., 2020).

Many climate indices share a common methodological architecture. They begin with a corpus of text, compare article content against a benchmark vocabulary or document, apply a quantitative procedure to identify relevant articles, and aggregate the resulting information into an index (Engle et al., 2020). Gentzkow et al. (2019) argue that successful text-based measures require vocabulary that closely corresponds to a clearly defined concept, validation against an external benchmark, and aggregation procedures that accurately capture the information contained within the underlying text. The credibility of a climate risk measure depends not on the amount of climate related text that is captured, but on whether its construction isolates the climate risk channel it claims to measure.

The Economic Policy Uncertainty Index (EPU) of Baker et al. (2016) established a key methodological template. By counting newspaper articles containing terms related to economic policy and uncertainty, the EPU demonstrated that text-based frequency counts could produce a credible measure of otherwise unobservable uncertainty. A direct attempt to introduce climate specificity within this framework is the Climate Policy Uncertainty (CPU) index of Gavriilidis (2021), developed by restricting the EPU keyword search to climate and energy policy terminology. The CPU index captures uncertainty surrounding government response to climate change transmitted

through credit markets via risk premia, lending standards, and credit supply (Facundo, 2026).

These measures narrow the vocabulary of policy-uncertainty without changing the underlying channel, thereby capturing transition risk without physical risk. The consequences of this conflation are visible in both US and European markets. Faccini et al. (2023) find that the climate policy factor carries a significant risk premium in US equity markets while physical risk factors remain largely unpriced. Similar results emerge in sovereign debt markets. Bua et al. (2024) construct separate transition and physical risk measures from Reuters archives and find that physical risk carries a distinct pricing signal in credit spreads. Anyfantaki et al. (2025) reach similar conclusions in global sovereign markets, showing that transition risk is reflected in sovereign bond yields.

2.3 Constructing a Credible Text-Based Index

Beyond the vocabulary choices, existing measures differ substantially in their construction. Validation, aggregation and robustness to construction are among the criteria on which existing physical-risk measures most often diverge. Engle et al. (2020) and Bua et al. (2024) employ near identical empirical approaches, yet they exhibit a correlation of just 0.01 (Jung and Hannaoui, 2024). Similarly, De Nard et al. (2024) report a correlation of only 0.17 between the New York Times and the Wall Street Journal's climate risk measures despite their comparable methodologies.

The literature has consequently placed increasing emphasis on validation. Baker et al. (2016) established the methodological framework by restricting vocabulary selection to terms that jointly referenced the economy, policy and uncertainty, and by validating their index against human-coded articles. Caldara and Iacoviello (2022) subsequently show that aggregating across multiple news sources reduces editorial bias and Type I errors while filtering high frequency false positives. Gavriilidis et al. (2026) further extend the validation standard by combining human coding with Large Language Model classification across 2,000 articles, achieving greater statistical significance. Validation practice has thus evolved from single method checks toward multi method, higher volume confirmation, establishing a rising bar against which new indices' construction choices must be benchmarked.

Aggregation choices present a further challenge. Frequency-based indices implicitly assume that every article contributes equal information, treating a wire service republication of a report as equivalent to the first article documenting the new extreme weather event. Loughran and McDonald (2011) illustrate that common terminology around the content dominates signal extraction and attenuates the informational value of frequency based measures. Similarly, Kalamara et al. (2020) show that retaining only first releases while discarding updates and

duplicates preserves the signal integrity by tracking new information rather than repetitive text.

Together, the literature recognises two recurring limitations: indices that combine physical and transition risk, and aggregation methods that treat repeated reporting as new information. The PCRA index addresses both limitations by restricting to physical risk content and incorporating novelty weighting that distinguishes new information from repeated coverage.

3 Index Construction

3.1 Corpus Construction

The PCRA index is constructed from newspaper articles retrieved through the Media Cloud platform (Roberts et al., 2021), an open-source archive that aggregates content from major national news organisations. The New York Times, USA Today, Reuters, CNN, CBS News, NBC News, and ABC News were chosen to reflect the national information environment of quality press journalism published between 1 January 2008 and 31 December 2025. The sample period includes economically important physical climate events, such as Hurricane Sandy (2012), Hurricane Harvey (2017), and Winter Storm Uri (2021).

Articles are retrieved from Media Cloud using a physical risk lexicon, defined in Appendix A.1, derived from the IPCC and TCFD physical risk taxonomies (IPCC, 2021; Task Force on Climate-related Financial Disclosures, 2017). The lexicon contains 112 terms, resolving to 50 distinct risk concepts. An article passes the topic filter if at least one lexicon term appears at least twice in the body of text, excluding articles in which physical risk vocabulary appears incidentally rather than substantively. Articles are additionally screened against a transition risk and climate finance exclusion lexicon (Appendix A.2), targeting transition policy vocabulary such as carbon tax, emissions trading, and net zero, following the physical-transition risk distinction of Faccini et al. (2023) and Bua et al. (2024). Any articles containing a term from the exclusion lexicon are dropped.

The pipeline is implemented in Python. Body text is extracted using trafilatura (Barbaresi, 2021), with newspaper3k (Ou-Yang, 2014) applied as a fallback where extraction is unsuccessful. Articles containing fewer than 50 words of extracted text are discarded. The text is then processed using spaCy's transformer pipeline (Honnibal and Montani, 2017), which performs tokenisation, lemmatisation, part-of-speech tagging, and named entity recognition. Articles are assigned to day t if timestamped before 16:00 Eastern Time, corresponding to the close of US Treasuries markets. Any articles published after this cutoff are assigned to day $t + 1$. This assignment is applied at the ingestion stage and is not modified downstream.

The initial corpus contains 54,400 articles. The corpus is subsequently passed through a secondary deduplication procedure to remove syndicated, republished, and near identical articles that would otherwise overstate attention on high coverage event days. Near duplicates are identified using Term Frequency and Inverse Document Frequency (TF-IDF) cosine similarity within a rolling 48 hour publication window. When a group of articles exceeds the similarity threshold, only one representative observation is retained from downstream scoring. The final corpus contains 41,922 distinct articles forming the basis of the PCRA index.

3.2 Article-Level Scoring

Each article in the final corpus receives a score reflecting its content of physical risk information. The score for article n published on day t, attributed to country k (United States), follows Arratia et al. (2021):

$$S_{n,t,k} \;=\; c(o) \cdot |l|^{-\alpha} \cdot \sum_{i \in \mathcal{L}} v_i \, s_i \, m_i \, \mathrm{rel}_{i,k} \tag{1}$$

where the summation runs over all tokens i in the article that belong to the physical risk lexicon $\mathcal{L}$, and $|l|$ is the article's token count. The outer term normalises by article length using the exponent $\alpha = 0.5$, following Loughran and McDonald (2011), which limits the influence of longer articles on the index. The source credibility weight $c(o)$ is set uniformly for all articles in the sample and remains fixed throughout.

Four multipliers determine each token's contribution. The severity weight $s_i \in \{1, 2, 3\}$ is calibrated ex ante against IPCC language categories: high impact vocabulary such as "catastrophic flooding" or "extreme heat" receives a weight of 3, medium impact vocabulary such as "storm damage" or "coastal erosion" receives 2, and baseline physical risk terms receive 1. The intensity modifier is defined as $m_i = 1 + \sum_j \mu_j$, which upweights tokens whose adjacent adjectives or adverbs amplify the risk signal. Each qualifying modifier j within a two token window contributes a increment μ_j defined in Appendix A.2 following Das and Chen (2007).

The valence shifter v_i captures whether the language affirms or suppresses the risk signal within a three-token window, following Polanyi and Zaenen (2006). Each term is given a scored number with negators set at $v_i = -1$, amplifiers set at $v_i = 1.5$, and de-amplifiers set at $v_i = 0.5$. The framework captures direct negation and amplification but does not explicitly model comparative downward framing (e.g. "fewer hurricanes than the previous decade"), which yields attenuated rather than negative scores.

The country relevance term, $\mathrm{rel}_{i,k} = 1/(1 + d_{i,k})$, discounts contributions from risk language that appears far from a US geographic entity mention in the token space, following Ding et al. (2008). $d_{i,k}$ is the token distance between the lexicon hit i and the nearest US geographic entity identified

by `spaCy`'s named entity recogniser. All scoring weights are assigned ex ante, registered prior to corpus construction following the pre-registration protocol recommended by Munafò et al. (2017) and applied in the uncertainty index literature by Hassan et al. (2019).

Valence and intensity act independently. Valence sets the sign while intensity scales the magnitude. The sentence "devastating floods caused widespread damage across the Midwest" scores highly. Valence is positive, severity is a 3 and intensity is elevated from "devastating" and "widespread" near a US geographic entity. By contrast, "analysts do not expect a devastating flood this season" receives a negative contribution regardless of the severity or intensity as the negator sets $v_i = -1$.

3.3 Novelty-Weighted Aggregation

Each article receives a novelty derived from dense sentence embeddings. For each article n published on day t, a document embedding $e_n \in \mathbb{R}^{384}$ is computed by encoding the article text with the `all-MiniLM-L6-v2` model in the sentence transformers framework of Reimers and Gurevych (2019). The novelty weight is

$$\omega_n \;=\; 1 - \max_{n' \in H_n} \cos(e_n,\, e_{n'}) \tag{2}$$

where H_n is the set of articles published in the prior seven days. The seven day baseline captures the typical lifecycle of a media event, during which repetitive coverage of a single episode is most concentrated. An article with no semantic precedent in the recent corpus receives a weight of $\omega_n = 1$ and contributes its score. Any near duplicate of recent coverage receives a weight ω_n approaching zero regardless of its severity or intensity score. As a robustness check, the window is also estimated empirically following Peress (2014), by computing average pairwise cosine similarity as a function of lag and identifying the lag at which mean similarity falls below 0.30. The daily score is the novelty-weighted mean across articles published on day t:

$$\bar{S}_{t,k} \;=\; \frac{\sum_n \omega_n \, S_{n,t,k}}{\sum_n \omega_n} \tag{3}$$

This produces a quality-weighted measure of information content rather than a volume measure of coverage. Doubling the number of near duplicate articles on a given day leaves $\bar{S}_{t,k}$ unchanged, since both numerator and denominator scale proportionally. The daily score will rise when genuinely new physical risk content enters the media environment and will remain stable when existing information is recirculated.

This distinguishes the PCRA index from article counting approaches, in which a second wire report on the same event counts as an independent observation, since repetition may itself raise public awareness even in the absence of new information. Section 5 tests this possibility directly.

3.4 Index Decomposition: SSI and CSP

The daily score $\bar{S}_{t,k}$ is decomposed into two components, the SSI and the CSP, following the convention of the news surprise literature of Beaudry and Portier (2006) and Barsky and Sims (2011). The SSI is an exponentially weighted moving average of the daily score:

$$\mathrm{SSI}_{t,k} \;=\; \lambda\,\mathrm{SSI}_{t-1,k} + (1-\lambda)\,\bar{S}_{t,k} \tag{4}$$

The baseline decay parameter is $\lambda = 0.977$, corresponding to a half-life of approximately 30 business days, following the RiskMetrics convention (RiskMetrics Group, 1996). Although no unique theoretical value exists, Section 5 shows the structural salience is robust to alternative half-lives of 15, 45 and 90 business days. Under the $\lambda = 0.977$ specification, coverage from the past 30 business days retains half its weight in today's SSI, while coverage from the last 90 business days has decayed to an eighth of its original weight.

The CSP isolates the unexpected component in two steps. First, a state adjusted score is formed by subtracting the previous day's SSI:

$$\tilde{S}_{t,k} \;=\; \bar{S}_{t,k} - \mathrm{SSI}_{t-1,k} \tag{5}$$

This removes the persistent component already known at the start of day t. Second, an autoregressive model AR(p) that includes day of week and month fixed effects is estimated from $\tilde{S}_{t,k}$, with p selected by the Bayesian Information Criterion (BIC) over the range $0 \le p \le 5$ to remove any remaining predictable serial structure, yielding the residual $\varepsilon_{t,k}$. The CSP is the standardised residual:

$$\mathrm{CSP}_{t,k} \;=\; \frac{\varepsilon_{t,k}}{\sigma_{\varepsilon_{t,k}}} \tag{6}$$

where $\sigma_{\varepsilon_{t,k}}$ is a trailing 252-rolling standard deviation using information available up to day t, accounting for time varying volatility in media attention. By construction the CSP is stationary, mean zero, and approximately orthogonal to $\mathrm{SSI}_{t-1,k}$. A positive CSP realisation on day t indicates that physical risk coverage exceeds the level implied by recent attention dynamics. No additional smoothing is applied to CSP because the measure is designed to capture day specific innovations in physical risk attention.

4 Properties of the PCRA Index

The PCRA index is benchmarked against the Emergency Events Database (EM-DAT) (Delforge et al., 2025). All US natural disaster entries across the meteorological, hydrological, climatological, and geophysical subtypes corresponding with the IPCC physical risk taxonomy are retained. Events affecting fewer than 100 people or causing less than 1 million US dollars in damage are excluded, leaving 417 events. These benchmark events provide an external basis for evaluating

the two components of the index.

The SSI is a unitless salience measure inherited from the underlying novelty-weighted article score and should be interpreted comparatively rather than absolutely. Higher values indicate a more persistent baseline level of attention to physical climate risk. By contrast, CSP is standardised by a trailing estimate of residual volatility and expressed in standard deviation units. A value +1 represents a one standard deviation positive innovation. Figure 1 presents the SSI and CSP components of the PCRA over the full sample. Their contrasting dynamics are clear: the SSI evolves smoothly, whereas CSP is characterised by short-lived spikes.

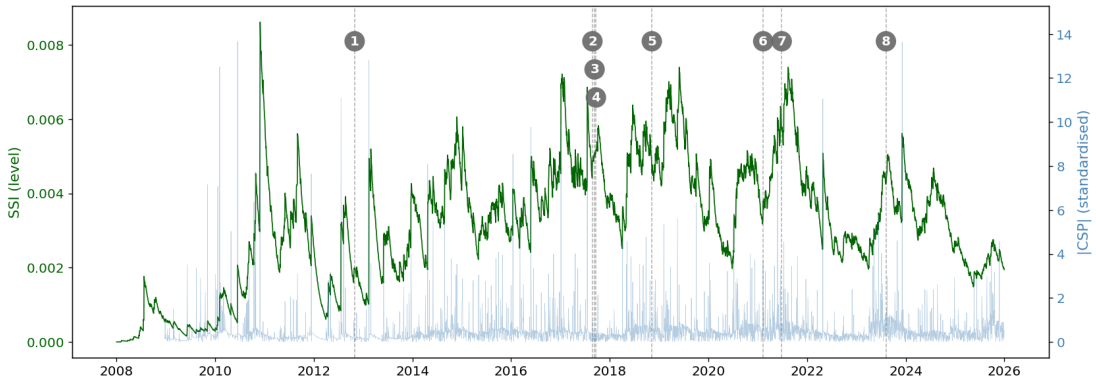


Figure 1: Daily SSI (left axis, green) and absolute daily CSP (right axis, blue bars). Labelled vertical lines mark eight canonical events: (1) Hurricane Sandy, (2) Hurricane Harvey, (3) Hurricane Irma, (4) Hurricane Maria, (5) Camp Fire, (6) Winter Storm Uri, (7) Pacific Northwestern Heat Dome, (8) Maui Wildfires.

.

4.1 The Structural Salience Index

The SSI is examined for evidence that salience rises with disaster frequency. Figure 1 shows the evolution of the SSI over the 2008-2025 sample. Rather than exhibiting a persistent upward trend, the SSI traces a non-monotone arc: rising through 2010-2011, declining, partially recovering during 2017-2019, and falling to its lowest sustained level by 2024-2025.

The SSI reaches its highest value in the sample on 29 November 2010 at 0.0086, a level never subsequently matched. The date is not attached to any of the sample's canonical named disasters. Instead, the highest peak follows a large winter storm system that crossed the western and north central United States between 22 and 25 November 2010. The storm brought a rare snowfall to Seattle and set a record for the city of Spokane's largest 24-hour November snowfall and, more consequentially, disrupting the heaviest Thanksgiving travel days (National Centers for Environmental Information, 2010).

The peak arose from ordinary winter storm coverage colliding with the highest attention travel week of the calendar, not from a single iconic disaster. It reflects the cumulative nature of salience, with the November 2010 peak stemming from an already elevated attention, not a stand

alone event. This global peak of the PCRA index predates the 2017 and 2018 hurricane seasons by over 6 years, suggesting that persistent salience accumulates across physical climate event types rather than within any single category.

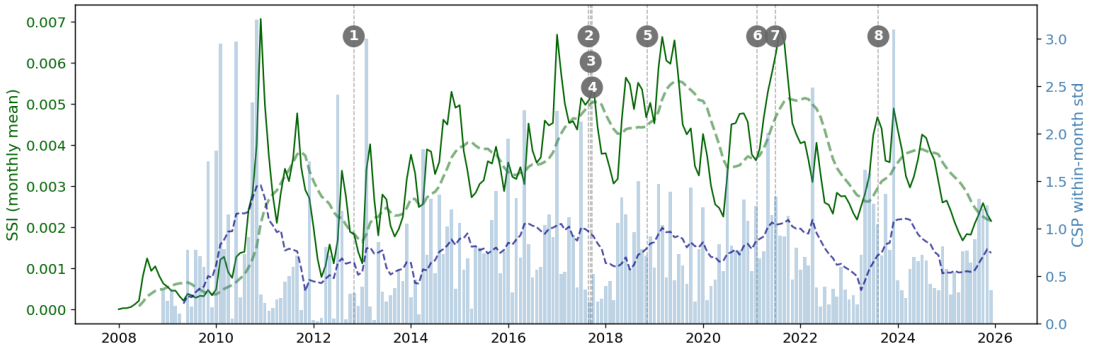


Figure 2: Monthly SSI mean (solid green) and its 12-month moving average (dashed green); monthly CSP mean (solid blue) and its 12-month moving average (dashed blue), with dark blue bars indicating within-month CSP standard deviation. Labelled vertical lines mark eight EM-DAT-recorded disaster events: (1) Hurricane Sandy, (2) Hurricane Harvey, (3) Hurricane Irma, (4) Hurricane Maria, (5) Camp Fire, (6) Winter Storm Uri, (7) Pacific Northwestern Heat Dome, (8) Maui Wildfires.

To assess whether this pattern extends beyond a single exceptional date, Figure 2 presents the monthly series. The twelve month moving average makes the two hump structure visible, at the same time the near constant flow of EM-DAT events across the same period confirms that this trajectory reflects information environment dynamics rather than changes in the underlying frequency of hazardous physical climate events.

4.2 The Climate Surprise Proxy

The CSP is examined for evidence that surprise spikes correspond mechanically to the largest disasters. Contrary to this expectation, the events most canonically associated with major US climate disasters are not the largest informational surprise in the sample. Figure 3 labels the largest positive CSP realisations across the full period with all of them being associated with winter storms, floods, and convective episodes. The concentration of extreme positive spikes in the 2008-2013 window reflects the lower SSI baseline prevailing in those years.

Anticipated events generate relatively small surprise responses because their information is incorporated before landfall, whereas unexpected events generate large CSP spikes. Hurricanes Sandy (1) and Harvey (2), together with the Camp Fire (5) are prominent SSI episodes but weak CSP events. Sandy (1) was extensively tracked before landfall, absorbing the informational surprise in the pre-event window. Hurricanes Irma (3) and Maria (4) arrived weeks after Harvey (2), into an information environment already primed by that coverage, and registered very weak CSP at landfall. These disasters were anticipated meaning much of their informational surprise was absorbed before impact.

Examining the benchmark events across the sample reveals three distinct patterns. Four hurricanes, Sandy (1), Harvey (2), Irma (3) and Maria (4), have CSP mass lying predominately in the pre-event window. Winter Storm Uri (6) and the Pacific Northwest Heat Dome (7) display opposite patterns with CSP spikes concentrated at and immediately after the event date. Others, like the Camp Fire (5) and Maui Wildfires (8), occupy an intermediate position, with CSP activity distributed across both sides of the event date, reflecting partial anticipation of elevated seasonal wildfire risk without precise event date prediction. This heterogeneity suggests that different classes of physical risk generate systematically different attention dynamics around the event date.

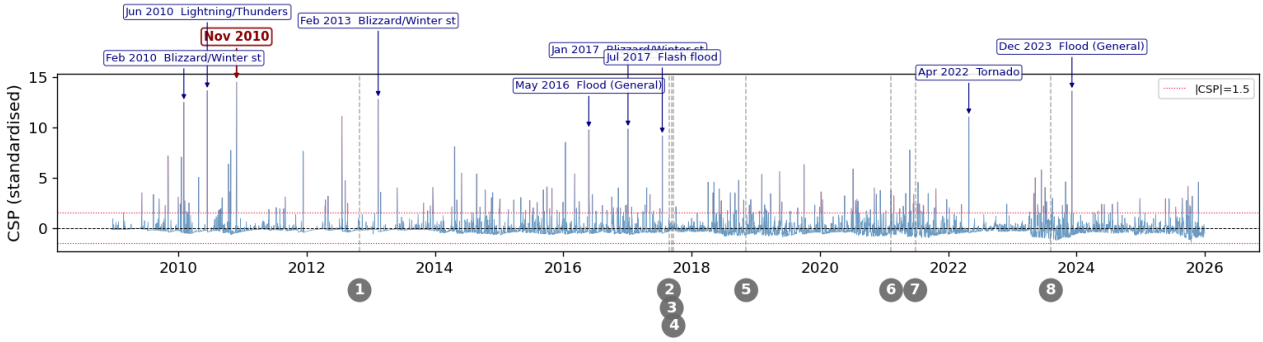


Figure 3: Daily CSP, 2008–2025. The largest positive realisations are each labelled with the nearest EM-DAT event. The largest ($|\text{CSP}| \approx 14.5$) occurs on 10 November 2010 during a winter storm episode. The pink dotted reference line is drawn at $|\text{CSP}| = 1.5$.

The concentration of the largest CSP realisations in the early part of the sample follows directly from this logic. On 2 February 2010, the prevailing SSI stood at 0.0012 and on 15 June 2010 at 0.0021. At those baseline levels, even moderate physical risk coverage produced large, standardised deviations. By January 2017, when a blizzard produced the eighth largest spike in the sample, the prevailing SSI was already 0.0069, and an equivalent article flow registered a substantially smaller surprise. The largest CSP realisations are concentrated in 2008-2013, not because physical climate risk events were more severe, but because the information environment was less accustomed to covering them.

4.3 Salience and Surprise Combined: the PCRA

Having distinguished salience from surprise, the analysis now examines their interaction over time. Rising structural salience raises the baseline level of climate risk attention, making genuine surprise increasingly rare. Figure 2 shows that the standard deviation within a month of daily CSP realisations moves in the opposite direction to the twelve month moving average of SSI over the full period. The within month CSP dispersion measure is highest in the 2008-2010 period, reaching a within month standard deviation of 3.20 in November 2010, and declines persistently thereafter, reaching its lowest sustained levels by 2023 and 2025.

This inverse relationship is not a mechanical artefact of the decomposition. It is a substantive finding about how the US information environment processes physical climate risk over time. As physical risk coverage has become an established feature of the media landscape, the threshold for an event to register as a genuine informational surprise has risen accordingly. A flood that constituted a major surprise in 2009 produces modest CSP realisations in 2023, not because floods have become less damaging, but because the expectation of climate coverage is now structurally embedded in the baseline against which the CSP is measured. This pattern is consistent with Ardia et al. (2023) and Gavriilidis et al. (2026), who find that deviations from expected climate coverage contain more economically relevant information than the level of coverage itself.

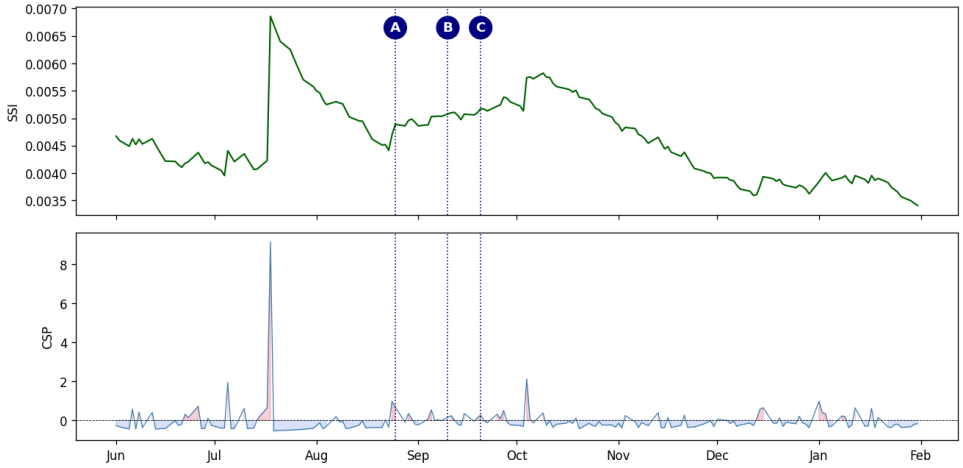


Figure 4: Daily SSI (upper panel) and CSP (lower panel) over the 2017 Atlantic hurricane season, June 2017 to January 2018. Red shading marks days on which CSP is positive and blue shading marks days on which CSP is negative. Dashed vertical lines mark the EM-DAT landfall dates for Hurricane Harvey (A, 25 August), Hurricane Irma (B, 10 September), and Hurricane Maria (C, 20 September).

Figure 4 focuses on the June 2017 to January 2018 window, which covers the 2017 Atlantic hurricane season. The largest CSP spike in the window, exceeding eight standardised units, occurs in late July, four weeks before Harvey's (A) landfall date, when early track model consensus first converged on a Texas landfall. As Harvey's coverage accumulates over the following weeks, SSI rises sharply. By the time Irma (B) makes landfall on 10 September and Maria (C) on 20 September, the SSI has already adjusted to a higher baseline, and CSP realisations at both event dates are negligible. The central finding is therefore that climate risk information is best understood relative to an evolving information environment. As climate risk becomes embedded in public discourse, larger shocks are required for an event to constitute new information.

5 Validation of the PCRA Index

5.1 Mechanical Robustness

The verification proceeds in three parts: mechanical replication of the exponentially weighted moving average recursion, the persistence of the SSI, and the innovation properties of CSP.

As a mechanical verification, λ is fitted back from the stored SSI series recovering $\lambda = 0.977159$ against the defined $\lambda = 0.977$ (Section 3.4). The maximum absolute deviation of 4.15×10^{-5} and mean absolute deviation of 3.65×10^{-6}, both within the tolerance threshold. As a robustness check, alternative SSI series are recalculated using 15, 45, and 90 business day half lives and they remain highly correlated with the baseline 30 business day series as seen in Appendix B.1.

The SSI is designed to measure the persistent baseline of attention to physical climate risk, rather than remove long run shifts. A cumulative sum (CUSUM) test for a structural break in the level of the series yields a maximum statistic of 20.104, well above the 5% critical boundary, indicating a real and sustained upward shift in media attention after 2014 (Figure 5).

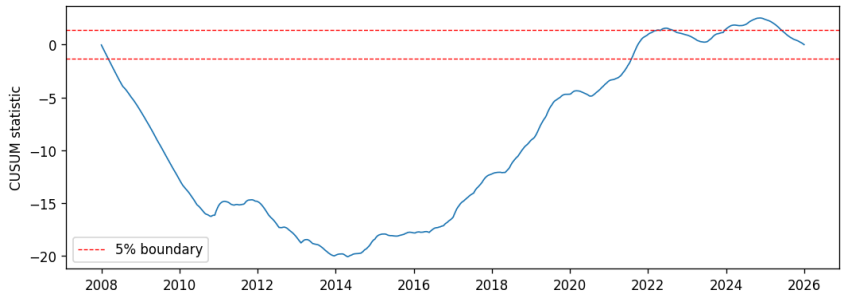


Figure 5: CUSUM-identified shift in media attention

The complementary construction requirement is that CSP isolates deviations from this persistent attention baseline. The standardised CSP meets its core construction targets, with a sample mean of -0.0028 and a standard deviation of 1.024. The construction diagnostics in Appendix B.1 confirm that CSP behaves as a stationary innovation series. By construction, CSP should be nearly orthogonal to lagged SSI, as confirmed by its Pearson correlation with SSI_{t-1} of -0.059, which remains within the pre-set $|\rho| < 0.10$ threshold.

The principal qualification concern is the non-linear dependence between CSP and lagged SSI. The distance correlation between the two is 0.188, exceeding the pre-set threshold of 0.15. CSP satisfies the linear orthogonality requirement but retains some non-linear dependence from the persistent attention state, suggesting that a large positive surprise is more likely to occur during periods of heightened baseline attention. Therefore, the dependence does not overturn the decomposition but qualifies the interpretation.

5.2 Layered Construction Validity

Having established the decomposition's mechanical validity, the analysis tests whether the subsequent construction choices improve the signal's informational content relative to simpler alternatives from the same article corpus.

Three specifications are derived from the same filtered article corpus, the differences across them

reflect construction choices rather than changes in sample composition. They form a nested progression from a naive measure to the full PCRA daily signal.

- *Article count*: the raw daily count of articles passing the lexicon filter, analogous to the Baker et al. (2016) EPU approach. This serves as the naive baseline.
- *Uniform-score signal*: the daily mean of article-level relevance scores with equal article weights, representing relevance scoring before novelty weighting is introduced.
- *Novelty-weighted signal*: the full daily PCRA signal defined in Equation (3).

Each specification is evaluated against two criteria. The first is serial dependence, measured by first-order autocorrelation $\phi(1)$. A lower $\phi(1)$ indicates that the series reflects new physical risk news rather than carrying forward momentum from prior reporting. The second is EM-DAT disaster coincidence. High attention days, defined as the top 10% of each standardised series, should occur near independently recorded US physical climate disasters more often than expected by chance. A day is counted as coincident if it falls within ±2 calendar days of at least one US EM-DAT event, with significance assessed against a 10,000 draw permutation null.

Table 2: Summary Statistics and EM-DAT Coincidence

Specification	$\phi(1)$	Excess kurtosis	Observed rate	Null mean	p-value
(i) Article count	0.632	26.1	28.1%	28.6%	0.604
(ii) S_{uniform}	0.089	68.1	35.7%	28.5%	0.001
(iii) S_{novelty} (full PCRA)	0.048	212.8	34.1%	28.5%	0.006

Table 2 shows that serial dependence falls monotonically across specifications. The uniform score signal cuts persistence by roughly 85%, while novelty weighting further reduces the remaining serial dependence.

Event validity improves in parallel. The raw article count does not coincide with EM-DAT disaster dates at a rate above chance ($p = 0.604$), confirming that article volume is noisy to recover the timing of physical risk events. Both scored specifications exceed the permutation null ($p = 0.001$ and $p = 0.006$), with the uniform score specification producing the strongest event alignment while the novelty weighted specification achieves lower serial dependence.

The distributional pattern is also consistent with the intended construction. Excess kurtosis rises from 26.1 for the article count to 212.8 for the novelty-weighted signal. Novelty weighting concentrates signal mass on fewer genuinely novel, high relevance days, which are event aligned rather than randomly distributed. The heavier tail, therefore, reflects a sharper concentration of signal around event-related construction choices, indicating that the PCRA behaves as an

information signal rather than a volume measure.

The pairwise correlations reinforce this interpretation. The article count is nearly uncorrelated with both scored series ($r = 0.028$ and $r = 0.002$), confirming that relevance scoring is not rescaling volume. This high correlation between the uniform score and novelty-weighted signals ($r = 0.943$) shows that novelty-weighting selectively refines the scored signal rather than replacing it. The results support the central construction choice that PCRA behaves as an information signal rather than a volume measure. These findings demonstrate that successive construction layers improve the statistical properties of the daily signal.

5.3 Event Validity

Event validity assesses whether the PCRA index captures real world physical risk events, rather than merely exhibiting internally consistent statistical behaviour. Testing whether its largest movements in the index coincide with independently recorded climate disasters. The test is applied to the final surprise component, CSP, to assess whether event coincidence survives the removal of the persistent attention component as relevance scoring and novelty-weighting have already been shown to improve the statistical properties.

Applying a $|\text{CSP}| \geq 1.5$ threshold to the 4,445 observation days identifies 216 surprise days, of which 73 (33.8%) fall within ±2 days of one of the 417 US EM-DAT events. Under a permutation null constructed from 10,000 random draws of equally many days the expected coincidence is 27.5% (59.4 matches, standard deviation 6.34), placing the observed count at the 98.4th percentile of the null and yielding a one sided p-value of 0.022.

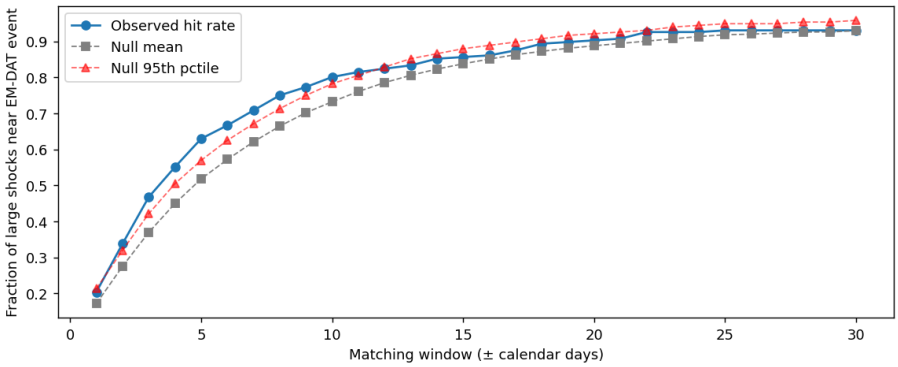


Figure 6: CSP & EM-DAT coincidence rate

To assess whether this result is specific to close event timing rather than an artefact of broad temporal overlap, Figure 6 compares the observed hit rate with the null distribution across matching windows from ±1 to ±30 calendar days. For short windows of ±1 to ±4 days, the observed hit rate exceeds the null 95th percentile, indicating that excess coincidence is concentrated near disaster dates. Beyond ±10, the observed rate converges toward the null mean

as the window size increases, reducing timing precision, supporting genuine news event timing over broad seasonal correlation.

Separating the EM-DAT events by type sharpens this interpretation. Acute events ($n = 369$), comprising sudden onset disaster such as storms, floods, and earthquakes, have a median offset of zero days to the nearest large CSP spike, consistent with same day news coverage of sudden disasters. Chronic events ($n = 48$), comprising slow onset conditions such as droughts and extreme temperature episodes, have a median offset of -3 days reflecting the weaker and less immediate timing alignment expected from slower moving physical risks. Overall, the large mean absolute offset of 29.6 days for acute events alongside the zero median demonstrates that large CSP spikes are sparse. The index captures precisely on the largest, most surprising events and stays quieter for the long tail of smaller disasters that do not breach the surprise threshold.

5.4 Human Coder Validity

The human coder audit tests whether the automated pipeline selects and scores articles that a human reader would classify as relevant to US physical climate risk. The audit uses 600 articles from 2014 to 2025 corpus selected based on the evidence in Section 5.1 indicating that sustained attention to physical climate risk becomes more visible after 2014. The sample includes 400 articles that passed the filter composed of 240 randomly selected articles, 80 drawn from each non-zero relevance score quartile and 80 with a score of zero. A further 200 rejected articles are selected to evaluate whether relevant physical risk articles were missed. The human coder's classification is established in Table 3.

Table 3: Human-Coder Audit: Variable Definitions

Variable	Type	Scale	Definition
Relevance	Binary	0 / 1	0 — Physical climate disasters, or direct impacts are absent. 1 — Physical climate disasters, or direct impacts are meaningful.
Intensity	Ordinal	1–5	Physical climate risk focus, from peripheral (1) to exclusive (5).
Attribution	Ordinal	1–3	US attribution, from none (1) to predominant/exclusive (3).

The audit defined five pre-specified thresholds. For binary physical risk classification, precision must reach at least 0.85 and recall at least 0.75, reflecting that false positives are more costly than missed articles since a noisy corpus corrupts every downstream index calculation, while missed articles can partially wash out in aggregation. Agreement with the human coder attempts to reach Cohen's $\kappa \geq 0.60$, the conventional threshold for "substantial" agreement (Landis and Koch, 1977), applied identically to the ordinal scoring function via weighted κ and Spearman's $\rho \geq 0.60$ against the human intensity ranking. These thresholds jointly indicate whether the constructed corpus is precise and complete to a level to support the analysis that follows.

Table 4: Human-Coder Audit Summary

Decision rule	Precision	Recall	Accuracy	F1	Cohen's κ
Physical-only rule	0.895	0.770	0.787	0.828	0.551
Combination rule	0.755	0.950	0.810	0.841	0.612

The human coder audit provides strong support for the PCRA pipeline's classification function and less support for this article-level intensity ranking function. Under the physical only rule, the pipeline meets the two central corpus construction thresholds: precision is 0.895 and recall is 0.77. This indicates that the corpus is not primarily driven by incidental keyword matches or generic policy coverage. The lower recall is consistent with the filter's conservative design, which prioritises a clean physical risk corpus over exhaustive coverage.

However, the classification does not meet the pre-specified chance corrected agreement target. Cohen's κ falls below the threshold at 0.551. Of the 256 articles the filter rejected, 92 (35.9%) were human-coded as relevant, while only 36 of 344 articles it passed (10.5%) were human coded as irrelevant. The shortfall is thus concentrated in missed relevant articles, not false positives consistent with precision and recall clearing the threshold.

Alternatively, classifying an article as relevant when it receives both a relevance score of 1 and a US attribution score of either 2 or 3 raises recall to 0.95 and Cohen's κ to 0.612, but at the cost of lowering precision to 0.755. The higher F1 score reflects a better balance between precision and recall, while the increase in accuracy indicates that a larger proportion of cases are classified correctly overall. This is conceived by admitting more of the same borderline content the physical only rule is designed to exclude. As the objective of the PCRA is to identify a non-repetitive dataset, this loss of precision is judged more costly than the gain in recall. The severity test produces a different conclusion from the classification tests, as it evaluates whether the score from equation (1) ranks articles in the same order of severity as a human coder would.

Among articles classified as physically relevant only (true positives), the Spearman correlation between the PCRA article level score and the human intensity rating is positive and statistically significant but weak ($\rho = 0.194$, $p < 0.001$, $n = 308$), falling below the pre-specified threshold of 0.60. When the comparison is extended to the combination rule, the Spearman correlation increases to 0.281. This increase demonstrates that most of the score's agreement with human judgement arises from correctly distinguishing relevant from irrelevant articles rather than from accurately ranking the severity of articles already identified as physically relevant. The evidence demonstrates that PCRA successfully identifies articles and that index spikes correspond to meaningful real world events.

5.5 Placebo Validity

The preceding validations establish that the index captures relevant content and aligns with independently recorded disasters, but it remains unclear whether this stems from the physical risk specific scoring framework itself or merely reflects the generic mechanics of the construction pipeline. To test this, the PCRA article level scoring function is replaced with the Loughran-McDonald negative sentiment lexicon, while all downstream steps are held fixed. If the pipeline generates the observed event alignment, the placebo should perform similarly.

The placebo index, denoted LM-CSP, replaces the PCRA article level relevance score with a length normalised count of Loughran-McDonald negative terms, as defined in Appendix A.3:

$$\tilde{s}_i^{\mathrm{LM}} = \frac{\sum_{z \in \mathcal{Z}_i} \mathbf{1}[z \in \mathcal{N}^-]}{|\mathcal{Z}_i|^{0.5}}, \tag{7}$$

where $\mathcal{Z}_i$ is the token sequence of article i, and $\mathcal{N}^-$ is the set of sentiment terms in the Loughran–McDonald dictionary, using the same length normalisation as the PCRA article level scoring.

This design isolates the role of physical risk specific content. The Loughran–McDonald lexicon is an approximate placebo since its terms measure general financial negativity rather than physical climate risk. If LM–CSP matches EM-DAT events as well as PCRA–CSP, the event validity result could reflect generic negativity or pipeline mechanics rather than the physical risk scoring function specifically.

The placebo signal has the same mechanical and statistical properties. LM–CSP is centred near zero, has a standard deviation close to one, and exhibits low first order autocorrelation. Furthermore, the BIC selected autocorrelation residualisation is AR(0), indicating no exploitable residual autocorrelation.

Table 5: Placebo comparison: PCRA-CSP versus LM-CSP

	PCRA-CSP	LM-CSP (placebo)
Event validity (EM-DAT, ±2-day window, 10,000 *draws)*		
Observed match rate	33.8%	24.7%
Permutation null mean	27.5%	28.5%
One-sided p-value	0.022	0.969
Construction properties		
First-order autocorr. $\phi(1)$	0.037	0.085
BIC-selected AR order	AR(0)	AR(0)

The event validity results differ sharply, as shown in Table 5. The two null means differ because each is calibrated to the number of surprise days in its own series. So each observed rate is

compared only against its corresponding null. PCRA–CSP spikes are clustered near EM-DAT events, where the general negative sentiment spikes are not. Thus, the disaster alignment results cannot be attributed solely to the pipeline's mechanics.

Correlations and principal component analysis reach the same conclusion. LM–CSP and PCRA–CSP are nearly uncorrelated, with Pearson correlation -0.026 ($p = 0.107$) and Spearman correlation -0.001 ($p = 0.937$). A two component principal component analysis on the standardised pair reveals an almost even variance split. The first principal component explains 51.3% of variance, showing PCRA captures a distinct signal rather than general text based sentiment.

5.6 Discriminant Validity

After establishing that the PCRA is specific to physical climate risk, evaluating whether it is an empirically distinct construct from existing uncertainty measures is a priority. Since a signal can be specific to physical risk content while still duplicating information captured elsewhere, the analysis examines the relationship between PCRA and established measures of uncertainty.

In theory, attention to physical climate risk should co-move only weakly with general policy uncertainty. The validation uses two monthly moments of the index: the calendar month average daily CSP (CSP_{mean}) and the largest absolute daily CSP within each month ($\text{CSP}_{\text{max abs}}$). The discriminant validity is assessed through four verifications: persistence, pairwise correlations, extreme-month overlap, and lead-lag dynamics.

The first verification is persistence. Monthly CSP_{mean} is centred near zero by construction with a mean of -0.002 and standard deviation of 0.213 and effectively no first order autocorrelation ($\phi(1) = -0.032$). By contrast, the CPU and EPU are persistent indices where CPU variants have first order autocorrelations between 0.780 and 0.873, while the EPU is even more persistent at $\phi(1) = 0.883$. Any relationship between CSP and the benchmark indices must arise from contemporaneous co-movement, not from shared persistence or common trending behaviour.

The second verification is pairwise correlation. Table 6 reports Pearson and Spearman correlations between CSP_{mean} and each benchmark series. Both CSP_{mean} and $\text{CSP}_{\text{max abs}}$ remain essentially orthogonal to the uncertainty benchmarks, with all absolute correlations below 0.11 and statistically indistinguishable from zero. By contrast, the CSP_{mean} and $\text{CSP}_{\text{max abs}}$ correlate strongly ($r = 0.701$), showing that the PCRA is internally coherent without overlapping the benchmarks.

Table 6: Pairwise correlations between monthly CSP and benchmark uncertainty indices

Pair	N	Pearson r	p-value	Spearman ρ	p-value
CSP_{mean} vs. CPU narrow	205	−0.103	0.141	−0.051	0.469
CSP_{mean} vs. CPU broad	205	−0.086	0.221	−0.015	0.833
CSP_{mean} vs. CPU Gavriilidis	202	−0.064	0.368	−0.036	0.611
CSP_{mean} vs. CPU news narrow	205	−0.081	0.250	−0.028	0.686
CSP_{mean} vs. CPU news broad	205	−0.063	0.368	−0.020	0.775
CSP_{mean} vs. CPU sentiment	145	−0.024	0.779	0.003	0.971
CSP_{mean} vs. CPU instrument	133	−0.076	0.383	−0.073	0.402
CSP_{mean} vs. CPU surprise	133	0.012	0.888	−0.016	0.860
CSP_{mean} vs. EPU	205	−0.003	0.968	−0.017	0.814

Figure 7 visualises the block structure where the benchmarks cluster together, while their association with the PCRA is negligible.

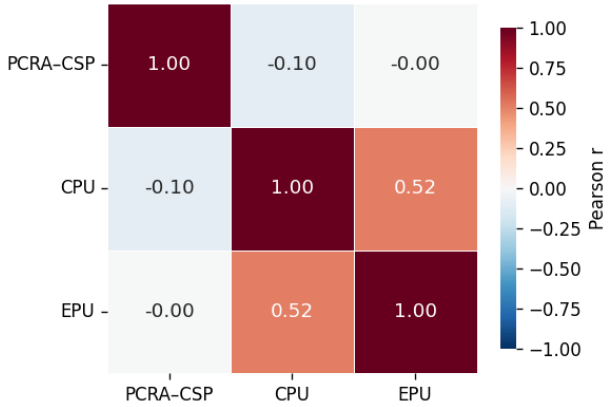


Figure 7: Pearson correlation matrix: PCRA–CSP, CPU, and EPU.

The third verification is extreme month overlap. The surprise overlap analysis tests whether the lack of correlation hides clustering in extreme months. Months are classified as high CSP or high CPU using the 75th percentile threshold for each series. Under independence, 13 of the 52 high CSP months would also be high CPU months. The observed overlap is a total of 11 months (21.2%), slightly below the null expectation. A Fisher exact test against the 2×2 classification gives $p = 0.47$, demonstrating no significant departure from independence. That is high physical risk attention months do not systematically coincide with high CPU months.

The final verification is lead–lag dynamics. Correlations between CSP_{mean} and CPU narrow are computed at lags -3 and $+3$ months. The correlations are insignificant at every horizon, with no asymmetry between leads and lags. This rules out a directional relation in which CSP either anticipates or follows CPU within a quarter of the year period. The contemporaneous correlation is the largest in absolute value but remains small and insignificant ($r = -0.103$, $p = 0.141$).

5.7 Principal Component Analysis

The last of these validations applies principal component analysis to the standardised combination of CSP_{mean}, CPU_{narrow}, and EPU. The tri-variate decomposition provides the clearest interpretation. The first principal component explains 50.9% of the variance and loads almost entirely on CPU_{narrow} and EPU (0.707 and 0.693 respectively), with negligible contributions from the CSP (-0.143). This is the policy and uncertainty factor proxying the transition risk. The second principal component explains 33.3% and loads almost entirely on CSP (0.980), identifying the physical risk factor. The third principal component explains 15.8% and captures the residual contrast between CPU and EPU after their shared component is removed. The CPU and EPU load jointly on shared policy and economic uncertainty component, while the PCRA loads almost entirely on its own separate factor.

6 Treasury-Curve Response Evidence

Physical climate events may affect sovereign borrowing conditions by reducing economic activity, increasing public expenditure, and damaging productive capital (Cevik and Jalles, 2020; Zenios, 2023). Studies of credit sensitive instruments have documented measurable pricing effects, including flood-risk premiums in US municipal bonds (Painter, 2020), event driven yield responses (Bernstein et al., 2019), and persistent increases in sovereign spreads associated with climate vulnerability (Cevik and Jalles, 2020). This evidence creates a natural prior that physical risk should also be priced into sovereign markets, consistent with the attention based frameworks of Engle et al. (2020) and Ardia et al. (2023). The PCRA decomposition allows this prior to be tested directly by purging persistent salience and isolating whether unexpected physical risk news move the US Treasury curve, instead of a gradual repricing of the climate concerns.

Since physical climate risks accumulate over long horizons and carry potential fiscal consequences, the central prediction is that any response should strengthen at longer maturities. Ten-year Treasury term premium and risk-neutral yield component, obtained from the Federal Reserve Bank of New York following the decomposition of Adrian et al. (2013), are used at a monthly frequency to test this prediction.

The evidence in this section documents a robust null instead. The CSP shows no detectable effect on the Treasury curve at any maturity or horizon, suggesting that pricing effects reported elsewhere may reflect salience driven correlation with other priced risk factors rather than a direct surprise driven repricing channel.

6.1 Baseline Local Projection Specification

Before testing this prediction, it is necessary to verify that CSP is not itself predicted by previous market movements. Three complementary tests, Granger-causality, a joint Wald test under Newey–West HAC inference, and distance correlation jointly indicate that past Treasury yield changes do not predict climate surprise. The full results are reported in Appendix B.3.

The predeterminedness evidence supports the use of local projections to test the maturity gradient hypothesis. These projections trace the yield response to a CSP surprise separately at each forecast horizon rather than assuming a single fixed effect. For each maturity $m \in \{2\text{Y}, 5\text{Y}, 10\text{Y}, 30\text{Y}\}$ and horizon h, the change in yield is regressed on the contemporaneous surprise:

$$y_{t+h}^{(m)} - y_{t-1}^{(m)} = \alpha_h^{(m)} + \beta_h^{(m)} \, \text{CSP}_t + \theta_h^{(m)} \, \text{SSI}_{t-1} + \nu_h^{(m)\prime} Z_t + \varepsilon_{t+h}^{(m)} \tag{8}$$

Here β_h^m traces the yield response to a one standard deviation CSP, and SSI_{t-1} enters lagged one day to avoid the mechanical contemporaneous correlation with CSP_t built into its construction. Z_t includes the Merrill Lynch Option Volatility Estimate (MOVE) index (representing the risk appetite), the CPU and EPU (representing the policy uncertainty and transition risk) and the Geopolitical Risk Index (GPR) (Caldara and Iacoviello, 2022). Together, these controls separate the response to new physical-risk information from broader risk appetite conditions, policy uncertainty, and persistent climate salience. This is consistent with the discriminant validity results in Section 5.6. Section 6.4 examines the stability of the estimate as SSI and each control are added or removed individually.

The Newey–West Standard errors are applied at every horizon to account for serial correlation induced by overlapping projection windows and are used uniformly across all specifications.

6.2 Main Result: A Robust Null

The estimates trace the predicted contemporaneous maturity ordering but provide no evidence of systematic Treasury repricing. The statistical evidence does not strengthen with either maturity or horizon. 0 of the 21 horizons are significant at the 5% level under the baseline specification, and only 1 of 21 under the fully controlled specification. Furthermore, all maturities fail to reject the joint null of no response across horizons (Table 7), and estimated coefficients are stable across the nested control specifications.

Table 7: Joint horizon tests across maturities

	2Y	5Y	10Y	30Y
χ^2 (p-value)	13.23 (0.900)	19.16 (0.575)	17.56 (0.677)	12.42 (0.928)
$\beta_0^{(m)}$	-2.45×10^{-6}	-1.46×10^{-6}	$+1.41 \times 10^{-6}$	$+2.19 \times 10^{-6}$

The contemporaneous coefficients trace the maturity ordering predicted by theory, yet this ordering is not statistically significant at any maturity. This contrasts with evidence from municipal bonds and sovereign spreads, where climate related information directly affects compensation for fiscal and default risk (Painter, 2020; Bernstein et al., 2019). At the national level of the Treasury market, any fiscal signal reflected in physical risk appears to be outweighed by monetary policy expectations, liquidity demand, and flight to quality dynamics.

A more direct test examines the 30Y–2Y yield spread. If CSP contains information about long horizon fiscal risk, this spread should widen following a positive surprise. Consistent with the maturity-gradient hypothesis, the estimated spread response is positive, but economically small and statistically insignificant ($\hat{\beta}_0 = 6.94 \times 10^{-6}$, $p = 0.45$). The predicted sign is carried but not the magnitude or precision needed for a meaningful pricing response.

The maturity gradient also does not survive beyond the contemporaneous horizon, clarifying why the initial ordering does not constitute persistent repricing. Seemingly Unrelated Regressions (SUR) estimates of the spread show that $\hat{\beta}_h^{(30Y)} - \hat{\beta}_h^{(2Y)}$ reverses sign after $h = 0$ and remains reversed at all subsequent horizons considered. The same steepening is recovered under both univariate Ordinary Least Squares (OLS) and SUR common sample restriction at both the daily and weekly frequencies, indicating that the reversal reflects a horizon effect rather than a sample composition artefact.

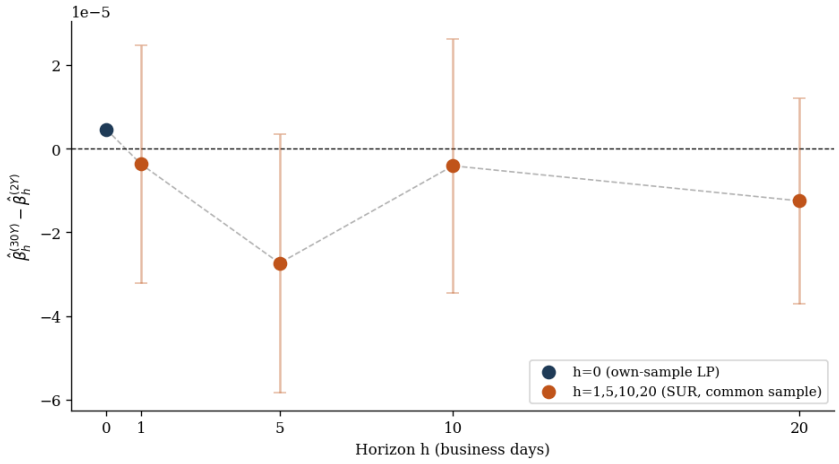


Figure 8: $\beta_h^{(30Y)} - \hat{\beta}_h^{(2Y)}$ plotted at $h = 0$ (own-sample) and $h = 1, 5, 10, 20$ (SUR), showing the sign reversal in the maturity gradient beyond the contemporaneous horizon.

6.3 Evaluating Alternative Explanations

The baseline null result opens a possible alternative interpretation. A change in CSP could raise long horizon fiscal compensation while simultaneously triggering safe haven demand, allowing the nominal yield to remain unchanged even though both components respond. To distinguish between these possibilities, the ten year yield is decomposed into its term-premium and risk

neutral components using the framework of Adrian et al. (2013), and each component is projected on CSP at horizons $h \in \{0, 1, 5\}$ in months, controlling for SSI_{t-1}, CPU and EPU.

Table 8: Term-premium decomposition: response of 10-year yield components to a change in CSP

	$h = 0$	$h = 1$	$h = 5$
Term premium (ACMTP10)	-5.1×10^{-2} (7.6×10^{-2})	-2.5×10^{-2} (1.0×10^{-1})	-9.4×10^{-3} (1.9×10^{-1})
Risk-neutral (ACMRNY10)	-4.3×10^{-2} (4.1×10^{-2})	-1.3×10^{-2} (5.8×10^{-2})	$+1.2 \times 10^{-2}$ (1.1×10^{-1})

Notes: Standard errors are in parentheses.

Table 8 shows a decline in the risk neutral component that occurs on impact, with the term premium moving in the same direction but neither support conventional significance ($p > 0.3$). This does not support cancellation, instead, the common negative sign implies reinforcement and would predict a contemporaneous decline in the ten-year yield, which Section 6.2 does not find reflecting an absence of response.

6.4 Specification choice

The 30Y–2Y spread regression is re-estimated using alternative combinations of CSP, SSI, and controls Z_t. The contemporaneous CSP coefficient remains virtually unchanged. The estimate shifts only from $\hat{\beta}_0 = 7.03 \times 10^{-6}$ in the CSP only model to $\hat{\beta}_0 = 6.94 \times 10^{-6}$ when the SSI is included alongside CSP and to $\hat{\beta}_0 = 6.93 \times 10^{-6}$ once the full control set is added. Additionally, by including the GPR (Caldara and Iacoviello, 2022) to the full model CSP leaves the coefficient essentially unchanged at $\hat{\beta}_0 = 6.99 \times 10^{-6}$ with a nested F-test value of $F = 0.274$, $p = 0.601$. Robustness of this kind converts an absence of evidence into evidence of absence.

Figure 9 plots the impulse response $\hat{\beta}_h^{(m)}$ from equation (8) across all four maturities under the full control specification. The coefficients are negative at shorter maturities and positive at longer maturities, reproducing the pattern established in Section 6.2. The paths oscillate around zero rather than developing into a persistent maturity dependent response. The shaded region is the 95% confidence interval based on Newey–West errors $(h + 1)$.

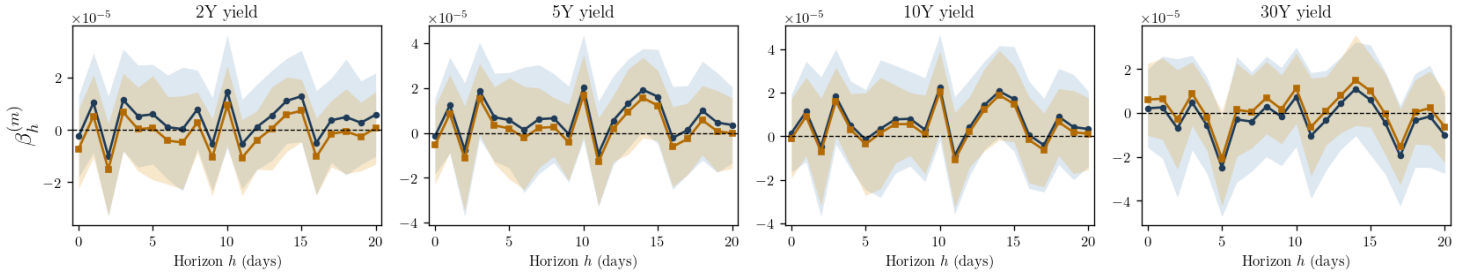


Figure 9: Treasury yield response to a change in CSP broken down into local projections across maturities.

Adding two lags of yield changes does not greatly alter the shape or horizon to horizon volatility of the impulse response paths. A delayed pricing interpretation is therefore ruled out. The

lagging CSP by one, three, and five trading days yields no significant response, with or without controls ($p \geq 0.108$ throughout). Altering the time frame to weekly forecasts reaches the same finding. Furthermore, the same finding holds when SSI is introduced individually across horizons.

6.5 Sign Counts, Events, Subsamples, and Asymmetry

The preceding results establish that the average Treasury response is weak and stable across specifications. Whether this average conceals economically important effects in particular events, subsamples, or non-linear specifications is examined through four investigations.

The ten-year maturity spans 147 estimated coefficients across the specification grid (Appendix C.4), of which four are significant at the 5% level against 7.4 expected by chance ($p = 0.939$). Extending to the full grid across all four maturities (588 coefficients) yields 18 significant estimates against 29.4 expected by chance ($p = 0.992$), concentrated at the five-year and thirty-year maturities (seven specifications each), with none at the two-year Treasury. They echo the sign reversal pattern already documented in the maturity gradient analysis in Figure 9 rather than signalling an independent effect. Furthermore, the spread between 30Y–2Y shows a one standard deviation CSP move only 0.072 basis points, an economically negligible magnitude.

The event study analysis around the ten largest CSP spikes similarly reveals no response concentrated near the event date. Testing the average response at each relative day in the event window produces only one significant estimate at the 10% level (day -10, $p = 0.096$), with no plausible economic interpretation given its position ten days ahead of the event day. The average response does not shift significantly between the ten days preceding and the ten days following a spike ($p = 0.863$), and the cumulative response over five trading days immediately following a CSP spike is likewise statistically indistinguishable from zero ($p = 0.797$). Subsample estimates reinforce the pattern, they remain directionally consistent across periods and decline after 2020.

Table 9: Asymmetry tests across maturities

Series	β_{pos} (p)	β_{neg} (p)	Ratio $\lvert\beta_{\text{neg}}/\beta_{\text{pos}}\rvert$	Wald p	n
30Y−2Y spread	9.03×10^{-6} (0.388)	-7.99×10^{-6} (0.838)	0.9×	0.693	3,916
2Y yield	4.94×10^{-6} (0.596)	-5.87×10^{-5} (0.177)	11.9×	0.179	4,176
5Y yield	5.20×10^{-6} (0.586)	-5.22×10^{-5} (0.258)	10.0×	0.247	4,176
10Y yield	1.10×10^{-5} (0.288)	-7.22×10^{-5} (0.115)	6.6×	0.095*	4,175
30Y yield	1.13×10^{-5} (0.288)	-6.58×10^{-5} (0.110)	5.8×	0.092*	3,916

Notes: β_{pos} and β_{neg} from a regression of the daily yield (or spread) change on the positive and negative parts of CSP separately. Wald test reports $H_0 : \beta_{\text{pos}} = \beta_{\text{neg}}$.

A null average response could still mask an asymmetric response if positive and negative surprise pull yields in opposite directions by similar magnitudes. Table 9 shows negative CSP generates a

larger coefficient than positive CSP at most maturities. The difference is rejected only at the 10% level for the ten-year and thirty-year maturities. It does not extend to the yield spread specification most directly motivated by the maturity gradient hypothesis. This asymmetry is suggestive, but too statistically weak and narrowly concentrated to alter the central interpretation.

Collectively a clear empirical boundary is defined, unexpected physical risk does not generate a persistent, maturity dependent, or economically meaningful response in benchmark US Treasury yields.

7 Industry-Level Equity Evidence

Physical climate risk is heterogeneous across industries (Sautner et al., 2023). Consequently, a market relevant physical risk surprise need not generate a uniform response in aggregate equity indices. Consistent with this, the Treasury analysis in Section 6 found no broad market-wide reaction to CSP news. This section therefore examines whether industry level equity returns respond differentially to climate risk surprise according to underlying physical exposure.

The analysis uses daily value weighted returns for Fama–French 12 and 49 industry portfolios (FF12 and FF49) from Kenneth French's data library (French, 2026) over 2008 to 2025, with FF49 serving as the primary specification. Firm level constituent data, used to assess whether industry level patterns extend to underlying firms, are obtained from Financial Modeling Prep (Financial Modeling Prep, 2026) and the HF Data Library (HF Data Library, 2026).

Physical exposure scores, ξ_i, are assigned ex ante using sectoral vulnerability rankings from the IPCC (2022). The qualitative low, medium, and high vulnerability categories are first mapped into numerical groups then refined by subdividing the high and low vulnerability tiers. This refinement is motivated by the substantial within category variation in physical surprise exposure documented by Sautner et al. (2023). The resulting measure is a fixed five point ordinal ranking that captures relative differences in physical vulnerability across industries rather than a calibrated measure of expected damages.

With industry and date fixed effects absorbing market wide shocks, identification rests on the cross-sectional ordering of physical exposure. The analysis proceeds from a common exposure slope to industry specific sensitivities, then to salience conditioned responses, firm level corroboration and economic magnitude.

7.1 Linear Exposure Rankings do not Explain the Cross-Section

Testing the uniform exposure pricing is conducted by determining whether a fixed exposure ranking produces a smooth return gradient across industries on the day of physical surprise. The

highest scores ($\xi \in \{4, 5\}$) are assigned to sectors such as Oil, Utilities, Agriculture, and physical infrastructure. The intermediate scores ($\xi \in \{2, 3\}$) are assigned to industries exposed through production facilities or supply chains. While the lowest scores ($\xi = 1$) are assigned to sectors with limited direct physical exposure, such as Business Equipment, Telecommunications, Health, and Finance.

If CSP captured physical risk news that investors priced uniformly across exposed industries, any industry with higher ξ_i should respond more negatively to an expected increase in CSP than industries with lower ξ_i. By examining a two-way fixed effects industry panel in which industry returns are related to the interaction between CSP and the fixed exposure score:

$$r_{i,t} = \alpha_i + \delta_t + \beta\,(\mathrm{CSP}_t \times \xi_i) + \theta\,(\mathrm{SSI}_{t-1} \times \xi_i) + \tau r_{i,t-1} + \varepsilon_{i,t}. \tag{9}$$

Here, $r_{i,t}$ is the daily industry return, α_i and δ_t denote industry and date fixed effects, respectively. The coefficient of interest is β, a negative estimate indicates a stronger negative response to CSP among more exposed industries. The lagged return term is included only as a robustness check. Date fixed effects absorb market wide surprises and common macro financial influences including the CPU and EPU, so identification comes from differential same day responses across exposure rankings. The interaction $\mathrm{SSI}_{t-1} \times \xi_i$ absorbs any persistent exposure related return gradient associated with structural climate salience.

Table 10: Industry-exposure panel: CSP $\times$ ξ_i interaction

	FF12	FF49
Daily		
$\hat{\beta}$	-3.7×10^{-7} (0.993)	-2.56×10^{-5} (0.390)
+ Lagged return $\hat{\beta}$	$+5.3 \times 10^{-7}$ (0.989)	-2.66×10^{-5} (0.372)
Industry-days (Baseline/+Lag)	51,420 / 51,408	209,965 / 209,916
Weekly		
$\hat{\beta}$	-5.4×10^{-4} (0.242)	-6.2×10^{-4} (0.080^{*})
Industry-weeks	10,680	43,610

Notes: p-values in parentheses.

The pooled exposure interaction provides little evidence of a stable linear gradient. The FF12 coefficient is essentially zero under the lagged-return specification, while FF49 estimates are directionally consistent with the exposure mechanism but statistically weak at daily frequency. When converted to a weekly frequency, the FF49 coefficient reaches marginal significance.

A minimum detectable effect analysis indicates that all specifications in Table 10 are underpowered relative to the estimated coefficients. For the daily baseline FF49 specification, the minimum detectable coefficient at 80% power is 8.3×10^{-5}, compared with an estimated coefficient of -2.6×10^{-5}, approximately one third of the magnitude that the design could reliably detect. The pooled result is therefore inconclusive as it cannot distinguish a genuine absence of pricing from heterogeneous industry responses that a common exposure slope cannot capture. The full

ex ante exposure assignment for all 49 industries is reported in Appendix D.1.

To address this, the common slope restriction is relaxed. Industry level CSP sensitivities are estimated with an ex ante exposure ranking. The exposure ordering hypothesis predicts that industries with higher exposure scores should exhibit more negative sensitivities.

7.2 Beta-Sort and Industry Level Sensitivities

The beta-sort evidence provides little support for a broad exposure gradient. Unlike the pooled panel in equation 9, which imposes a common slope on the $\text{CSP}_t \times \xi_i$ interaction, the beta-sort allows industry specific CSP sensitivities ($\hat{\beta}_i^{\text{CSP}}$) to be estimated separately from each of the 49 industries before relating them to ξ_i. Industries with higher exposure scores tend to exhibit more negative CSP, but the relationship is weak and statistically insignificant across specifications.

Table 11 shows the exposure slope remains negative and directionally consistent across specifications. However, neither the estimated slopes nor the corresponding Spearman correlations reach conventional significance levels, providing little evidence for exposure ordering.

Table 11: Cross-sectional beta sort: industry CSP sensitivities on physical exposure ξ_i

Specification	Slope $\hat{b}$	p_b	Spearman ρ	p_ρ
CSP only	-2.55×10^{-5}	0.248	-0.199	0.171
$+\,\text{SSI}_{t-1}$	-2.62×10^{-5}	0.231	-0.195	0.179
$+\,\text{SSI}_{t-1}$ + FF5	-2.37×10^{-5}	0.260	-0.157	0.282

Notes: $\hat{b}$ is the cross-sectional slope of $\hat{\beta}_i^{\text{CSP}}$ on normalised exposure ξ_i. p_b, p_ρ are the p-values.

Table 12 provides a descriptive check on the beta-sort results of the top 4 most extreme industries, benchmarked against the full cross section: Beer, Soda, and Smoke are significant at the 5% level ($p = 0.006$, $p = 0.012$, and $p = 0.044$ respectively). Extending to the 10% level Agriculture is additionally significant ($p = 0.080$). The remaining coefficients are not statistically significant, reinforcing the finding that responses vary across industries rather than aligning along a common exposure gradient. The full results are reported in Appendix D.3.

Table 12: Selected industry-level CSP betas: tails of the FF49 beta distribution

Most negative CSP *sensitivities*						*Most positive* CSP *sensitivities*					
Industry	ξ_i	$\hat{\beta}_i^{\text{CSP}}$	SE	t	p	Industry	ξ_i	$\hat{\beta}_i^{\text{CSP}}$	SE	t	p
Gold	3	-4.06	2.87	-1.41	0.158	Construction	4	$+2.66$	1.67	1.59	0.111
Soda	3	-3.44	1.36	-2.52	0.012	Mines	4	$+2.39$	1.68	1.42	0.154
Fabricated Products	2	-2.70	2.63	-1.03	0.304	Toys	1	$+2.06$	1.53	1.34	0.180
Beer	3	-2.60	0.95	-2.75	0.006	Business Services	1	$+1.35$	0.66	2.03	0.042

Notes: $\hat{\beta}_i^{\text{CSP}}$ and SE reported $\times 10^{-4}$. $n = 4{,}285$ industry-days per industry.

Both tails of the industry beta distribution illustrate the point. Some industries with plausible

physical supply-chain exposure load negatively on CSP, whereas Mines and Construction load positively, consistent with offsetting supply disruption, commodity pricing, or reconstruction of the demand channels. These economically interpretable responses prevent the cross-section from aligning along a clean monotonic exposure gradient. Instead, the results indicate a more selective mechanism in which physical risk surprise are concentrated in particular industries.

7.3 Salience Conditioned Response

The central empirical claim of the PCRA decomposition is that a given CSP carries greater pricing when SSI is high. If investors are more attentive during high salience periods, their responses to new surprise produce larger and faster market effects.

Each industry's daily return is regressed on CSP, its interaction with an indicator for high SSI periods defined as a lagged, SSI_{t-1} exceeding its 75th percentile. This regime specification is estimated separately for each industry, so the pooled coefficients β and θ in equation 9 become industry specific β_i, θ_i as seen in equation 10 where ξ_i no longer enters directly since it is fixed within each industry's own regression. A new coefficient γ_i is introduced as an additional interaction parameter. The estimated interaction coefficient $\hat{\gamma}_i$ measures the incremental CSP response during high salience periods. A negative estimate indicates amplified return losses when CSP arrives while attention is already elevated.

$$r_{i,t} = \alpha_i + \beta_i \,\text{CSP}_t + \gamma_i \,(\text{CSP}_t \times \mathbf{1}\,[\text{SSI}_{t-1} > p75]) + \theta_i \,\text{SSI}_{t-1} + \varepsilon_{i,t}. \tag{10}$$

Before restricting to the high-exposure tier, a cross sectional regression of $\hat{\gamma}_i$ on ξ is first estimated across all 49 industries. This cross section demonstrates a significant slope of -3.30×10^{-4} ($p < 0.0001$, $R^2 = 0.156$), with a Spearman rank correlation of -0.482 ($p = 0.0005$). The full sample result is more than a simple exposure gradient, elevated salience reallocates return sensitivity across the exposure spectrum. Table 13 decomposes this pattern by tier.

Table 13: SSI regime amplification by exposure tier, all 49 industries

Tier	Mean $\hat{\gamma}$	Amplify	Individually significant
High exposure ($\xi \geq 4$)	-1.97×10^{-4}	10/13	Food ($p = 0.037$), Utilities ($p = 0.011$)
Mid exposure ($\xi = 2, 3$)	-0.08×10^{-4}	12/26	None
Low exposure ($\xi = 1$)	$+4.06 \times 10^{-4}$	0/10	Guns (0.046), MedEq (0.081), Softw (0.088), Books (0.092)

During high SSI, a tiered pattern emerges with high exposure industries have a negative average regime increment, mid exposure near zero and low exposure the opposite direction. Although the Kruskal-Wallis test does not reject equality across tiers ($p = 0.174$), Table 14 indicates this high exposure average is concentrated rather than shared. 10 of the 13 industries classified as high exposure show a negative coefficient, but only Food and Utilities reach individual significance.

Table 14: SSI regime amplification: high-exposure industries ($\xi \geq 4$)

Amplifying				*Amplifying (cont.)*				*Reversing (positive $\hat{\gamma}_i$)*			
Industry	ξ_i	$\hat{\gamma}_i$	p	Industry	ξ_i	$\hat{\gamma}_i$	p	Industry	ξ_i	$\hat{\gamma}_i$	p
Food	4	-5.65	0.037^{**}	Shipping	4	-4.12	0.464	Agriculture	5	$+0.229$	0.930
Insurance	4	-5.04	0.146	Transport	4	-3.96	0.328	Mines	4	$+5.61$	0.365
Real Estate	5	-4.47	0.145	Paper	4	-3.16	0.378	Coal	4	$+8.75$	0.471
Utilities	5	-4.45	0.011^{**}	Chemicals	4	-2.62	0.519				
Oil	5	-4.21	0.200	Construction	4	-2.54	0.628				

Notes: $\hat{\gamma}_i$ reported $\times 10^{-4}$.

The significance coefficients identified are consistent with hypothesised rotation, but they represent only two industries. Table 15 shows a broader verification to isolate every industry that clears significance. Across all three tiers, the two significant high exposure industries amplify while all four significant low exposure industries reverse, consistent with rotation toward less physically exposed sectors when salience is elevated.

Table 15: All individually significant industries, SSI regime interaction

Amplifying				*Reversing*				*Reversing (cont.)*			
Industry	ξ_i	$\hat{\gamma}_i$	p	Industry	ξ_i	$\hat{\gamma}_i$	p	Industry	ξ_i	$\hat{\gamma}_i$	p
Food	4	-5.65	0.037	Guns	1	$+5.92$	0.046	Software	1	$+5.80$	0.088
Utilities	5	-4.45	0.011	Medical Eq.	1	$+5.06$	0.081	Books	1	$+7.32$	0.092

Notes: $\hat{\gamma}_i$ reported $\times 10^{-4}$.

The individual significance of the industries is weak revealing only one of the reversing (Guns, $p = 0.046$) industries clears the 5% significance level. This cannot distinguish the salience conditioned channel from a false discovery, instead the evidence lies in the broader cross-sectional pattern.

The regression of $\hat{\gamma}_i$ on ξ_i restricted to the high exposure industries does not reach significance (-1.81×10^{-4}, $p = 0.416$, $R^2 = 0.040$, $n = 13$), but it is nonetheless still consistent with the tier level pattern. The full sample and restricted regressions are complementary rather than competing in this respect. The former identifies the cross tier rotation as the dominant pattern in the data, while the tier level breakdown, the Food and Utilities coefficients, and their firm level corroboration in Section 7.4 provide the direct evidence for amplification with exposed industries. The emergence of conditionality suggests that the exposure alone is insufficient to identify where physical risk is priced. Instead, the pricing response appears concentrated in particular information environments.

7.4 Firm Level Corroboration of the Industry Results

The industry level amplification is concentrated in specific sectors. To establish that this effect is not an artefact of portfolio construction, the daily returns for the constituent firms are examined directly using equation (10).

The firm level regressions for Food and Utility demonstrate that the regime coefficient becomes more significant when the analysis moves to firm panels. If the industry level amplification is driven by portfolio construction, then disaggregation should weaken the signal due to the introduction of firm specific noise. Instead, the opposite occurs. The Food panel yields $\hat{\gamma} = -0.084$ ($p = 0.003$) and Utilities yields $\hat{\gamma} = -0.092$ ($p = 0.005$). Both are statistically significant and both exceed the minimum detectable effect threshold demonstrating a true finding.

Figure 10 makes the pattern visible. All Food and Utility firms have a negative regime coefficient, although not every individual estimate reaches conventional significance. Specifically, 7 of 14 Food firms and 25 of 34 Utility firms are significant at the 10% level, and a smaller portion of only 4 and 15 respectively at the 5% level. Full firm by firm regime coefficients for both industries are reported in Appendix D.5. Some non-significant estimates carry wider standard errors, likely reflecting shorter trading histories and greater idiosyncratic volatility among smaller, more recently listed firms.

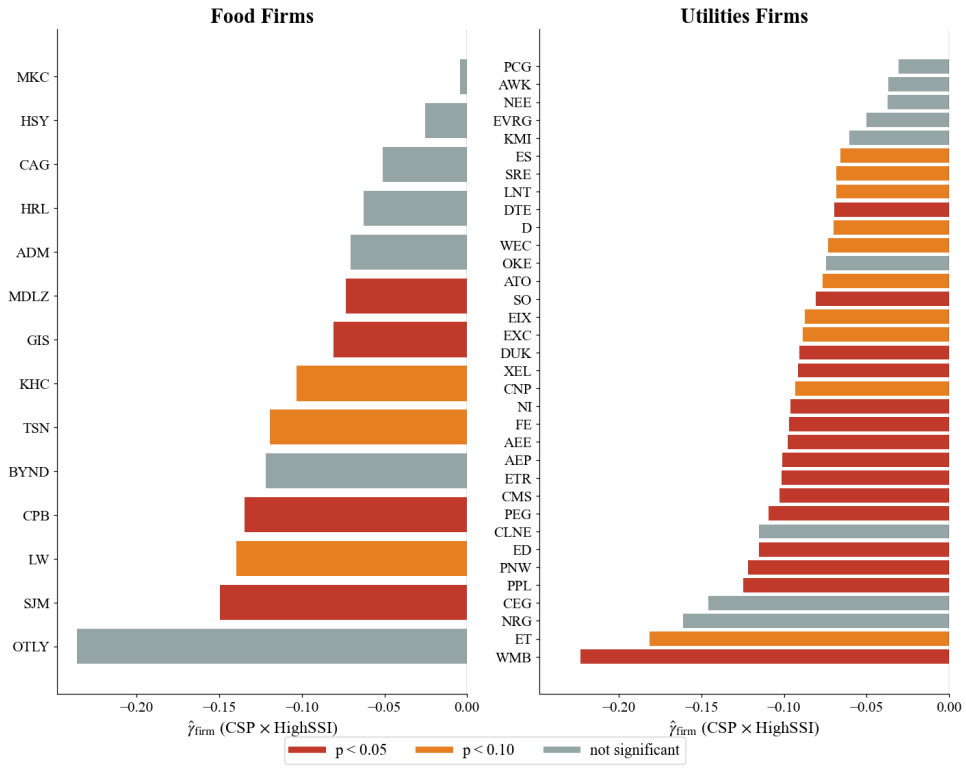


Figure 10: Firm-level regime coefficients for Food and Utilities constituents, sorted by magnitude and coloured by significance tier.

The unanimous sign consistency is also assessed formally. Under the null that each firm's coefficient is equally probable to be positive and negative, a two sided binomial sign test shows that the probability of observing all fourteen Food firms with the same negative sign is $Prob = 0.0001$, with the corresponding probability for all 34 Utility firms smaller still. Both probabilities fall below conventional significance demonstrating that the firm level sign consistency is unlikely to reflect sample variation alone. As a further test of specificity, the same regime specification is estimated by pooling firms across the six industries with individually significant regime coefficients $\hat{\gamma}_i$ in Table 15. This pooled cross-industry firm panel produces no significant effect ($p = 0.986$), meaning the salience conditioned response does not generically emerge from firm-level data, but remains tied to the within industry structure identified in Section 7.3.

7.5 Economic Magnitude

The panel regressions establish that CSP co-moves more strongly with exposed industry returns when structural salience is elevated, but statistical detectability is not the same as economic exploitability. Expressed as a share of daily return volatility, the high SSI effect is consistently larger than the low SSI share as illustrated in Figure 11. Utilities are the largest in both periods, reaching a 3% share during high SSI periods.

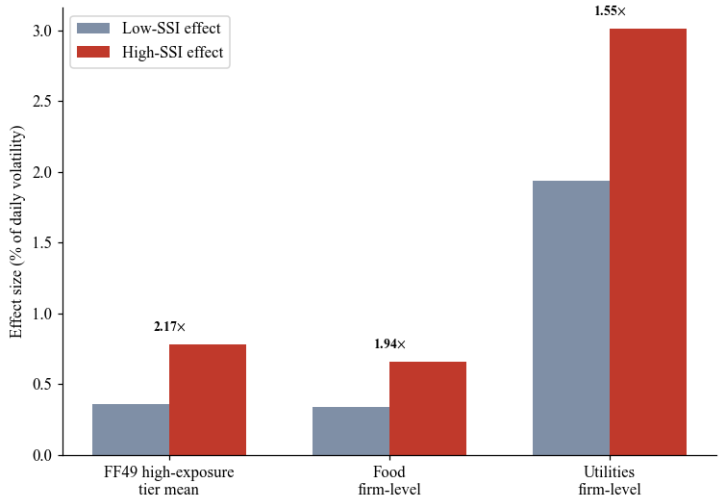


Figure 11: Economic effect size by SSI regime, expressed as a percentage of daily return volatility.

To complement the volatility normalised estimates, a model free difference-in-difference test independently compares the return spread between the most positive and negative CSP terciles across SSI regimes. Table 16 show a stronger negative return spread during high attention periods with the FF49 industries estimate being significant at the 1% level, whereas the Food and Utility firm level estimates are significant at the 10% level.

The economic interpretation is therefore not that CSP generates large daily return movements, but that it is dependent on how much physical climate risk attention is being received at the

time in both diverse industry portfolios and at the sector level.

Table 16: Difference-in-differences confirmation: CSP tercile return spread, low- versus high-SSI regime

	Difference-in-differences	t	p
FF49 industry panel	−0.00046%	−2.65	0.008
Food firm-level	−0.261%	−1.79	0.074
Utilities firm-level	−0.044%	−1.72	0.086

Notes: A negative difference indicates a more negative CSP-return relationship in the high-SSI regime.

However, this state dependence is not exploitable for large gains. Following Faccini et al. (2023), CSP is tested directly for pricing in the cross section of the 138 constituent firms drawn from the six industries in Table 15. Firms sorted monthly into quintiles by their trailing 63 day CSP beta (Q5–Q1) yield spread alphas of 0.19% to 0.32% per month, none distinguishable from zero. Conditioning on salience by restricting to high SSI periods yields -0.07% and -0.22% per month, again indistinguishable from zero at daily, weekly and monthly frequencies. Sample sizes are sufficiently large that the absence of significance is not an underpowered result (Appendix D.6).

Food and Utility portfolios also produce no significant alphas under the same specifications. Food alpha's range from 1.87% to 2.44% per month and Utility's from 0.05% to 0.12% per month, both statistically indistinguishable from zero. The findings demonstrate that even with a real physical climate risk signal present, the signal does not convert into an exploitable strategy.

8 Conclusion

Physical climate is increasingly recognised as a financial risk, yet existing text-based indices do not isolate it as a distinct channel. Most indices conflate physical and transition risks, treat repeated coverage as new information, and collapse persistent salience and high-frequency surprise into a single series. This thesis introduces the Physical Climate Risk Attention (PCRA) index, a newspaper-based high frequency measure of physical climate risk constructed from national outlets in the United States for the period 2008 to 2025. The PCRA succeeds in isolating a distinct physical climate risk information channel, although the financial effects of this information are not uniform across asset classes or consistent across information environments.

Validation confirms the PCRA as a distinct physical climate risk information channel, showing only minimal overlap with existing indices CPU and EPU. Correlations stay weak, extreme months do not overlap more than chance implies, lead-lag dynamics show no relationship, and the PCRA loads onto its own principal component. The decomposition into the SSI and CSP extends this distinctiveness, separating persistent attention from unexpected information arrival to demonstrate that the risk surprise's information content depends on the prevailing salience

environment.

This conditionality extends to financial markets. Although CSP does not generate a detectable maturity gradient response in Treasury yields and unconditional equity evidence remains limited, pricing effects become evident when physical exposure and salience are considered together. Among highly exposed industries, specifically Food and Utilities, responses to climate risk surprise intensify during periods of elevated structural salience. While the least exposed industries exhibit opposite movements, consistent with a salience driven rotation across the exposure spectrum.

The distinguishing evidence emerges at the firm level within the Food and Utilities sectors, where the industry level amplification pattern is replicated across constituent firms. This pattern is reflected in unanimous sign consistency across all 14 Food and 34 Utilities firms examined, with half of Food firms (7 out of 14) and nearly three-quarters of Utilities firms (25 out of 34) are individually significant at the 10% level. The effect is economically meaningful, with a magnitude of one and a half to just over two times greater in high salience periods compared to low salience periods. However, this effect does not correspond to identifiable reactions around individual disaster events. Physical climate risk appears to be incorporated into equity prices, but only through a selective salience dependent channel rather than an exploitable mispricing.

8.1 Discussion and Extensions

While the results establish physical climate risk as a distinct information channel, they also show that its financial effects are heterogeneous. Aggregate financial markets may therefore conceal economically important cross sectional response leading to three directions for future research.

First, more robust measures of firm-level exposure could enhance cross-sectional identification. Sector level IPCC ranking are insufficient to capture the substantial heterogeneity present within sector industries. An extension would replace the static, ordinal ξ_i measure with a continuous, firm specific exposure metric derived from geographic, supply-chain or earnings call data. This tests whether firms that emphasise physical risks more strongly exhibit corresponding greater CSP sensitivity. Such measures would substitute management's revealed assessment of physical risk for a sector assigned ranking and help explain the weak unconditional exposure gradient.

Second, the range of assets and instruments examined could be expanded. Since benchmark sovereign yields do not provide the most informative context for studying physical climate risk attention, future applications of PCRA should prioritise assets in which physical exposure is more directly reflected. Sovereign and corporate credit spreads, municipal bonds, and insurance linked securities represent alternatives to broad market instruments. Testing the PCRA against credit default swaps, rather than benchmark yields, would isolate the default risk component of

sovereign pricing from the liquidity and monetary policy influences that dominate risk free rates. The contrast between economically significant sector level equity response and the absence of a tradable return premium suggest that physical climate risk is primarily priced as tail risk rather than as a first moment return effect. An examination of option implied downside protection in sectors during high salience periods would provide a direct test of this mechanism.

Third, future research could enhance the precision and generalisability of the PCRA framework itself. Calibrating the severity score against observed disaster losses would strengthen its interpretation as a measure of physical climate risk magnitude rather than linguistic intensity alone. Comparing variations in the PCRA for other global economies would determine whether relationships between physical risk, salience and financial pricing generalises beyond the US. Implementing a multi-coder benchmark would further strengthen generalisability standard, providing a clearer assessment of classification performance.

References

Adrian, T., Crump, R. K., and Moench, E. (2013). Pricing the term structure with linear regressions. *Journal of Financial Economics*, 110(1):110–138.

Anyfantaki, S., Blix Grimaldi, M., Madeira, C., Malovaná, S., and Papadopoulos, G. (2025). Decoding climate-related risks in sovereign bond pricing: A global perspective. BIS Working Paper 1275, Bank for International Settlements.

Ardia, D., Bluteau, K., Boudt, K., and Inghelbrecht, K. (2023). Climate change concerns and the performance of green vs. brown stocks. *Management Science*, 69(12):7607–7632.

Arratia, A., Avalos, G., Cabaña, A., Duarte-López, A., and Renedo-Mirambell, M. (2021). Sentiment analysis of financial news: Mechanics and statistics. In Consoli, S., Reforgiato Recupero, D., and Saisana, M., editors, *Data Science for Economics and Finance*, pages 195–215. Springer.

Baker, S. R., Bloom, N., and Davis, S. J. (2016). Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4):1593–1636.

Barbaresi, A. (2021). Trafilatura: A web scraping library and command-line tool for text discovery and extraction. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics: System Demonstrations*, pages 122–131.

Barsky, R. B. and Sims, E. R. (2011). News shocks and business cycles. *Journal of Monetary Economics*, 58(3):273–289.

Beaudry, P. and Portier, F. (2006). Stock prices, news, and economic fluctuations. *American Economic Review*, 96(4):1293–1307.

Bernstein, A., Gustafson, M. T., and Lewis, R. (2019). Disaster on the horizon: The price effect of sea level rise. *Journal of Financial Economics*, 134(2):253–272.

Bua, G., Kapp, D., Ramella, F., and Rognone, L. (2024). Transition versus physical climate risk pricing in european financial markets: A text-based approach. ECB Working Paper 2677, European Central Bank.

Caldara, D. and Iacoviello, M. (2022). Measuring geopolitical risk. *American Economic Review*, 112(4):1194–1225.

Capelle-Blancard, G., Crifo, P., Diaye, M.-A., Oueghlissi, R., and Scholtens, B. (2019). Sovereign bond yield spreads and sustainability: An empirical analysis of OECD countries. *Journal of Banking & Finance*, 98:156–169.

Cevik, S. and Jalles, J. a. T. (2020). This changes everything: Climate shocks and sovereign bonds. IMF Working Paper 20/79, International Monetary Fund.

Das, S. R. and Chen, M. Y. (2007). Yahoo! for amazon: Sentiment extraction from small talk on the web. *Management Science*, 53(9):1375–1388.

De Nard, G., Engle, R. F., and Kelly, B. (2024). Factor-mimicking portfolios for climate risk. *Financial Analysts Journal*, 80(3):37–58.

Delforge, D., Wathelet, V., Below, R., Lanfredi Sofia, C., Tonnelier, M., van Loenhout, J. A. F., and Speybroeck, N. (2025). EM-DAT: The emergency events database. *International Journal of Disaster Risk Reduction*, 124:105509.

Dell, M., Jones, B. F., and Olken, B. A. (2012). Temperature shocks and economic growth: Evidence from the last half century. *American Economic Journal: Macroeconomics*, 4(3):66–95.

Ding, X., Liu, B., and Yu, P. S. (2008). A holistic lexicon-based approach to opinion mining. In *Proceedings of the 2008 International Conference on Web Search and Data Mining*, pages 231–240.

Engle, R. F., Giglio, S., Kelly, B., Lee, H., and Stroebel, J. (2020). Hedging climate change news. *Review of Financial Studies*, 33(3):1184–1216.

Faccini, R., Matin, R., and Skiadopoulos, G. (2023). Dissecting climate risks: Are they reflected in stock prices? *Journal of Banking and Finance*, 155:106948.

Facundo, L. M. (2026). Climate policy uncertainty and macroeconomic dynamics: Financial amplification and state-dependent effects. *Economic Modelling*, 158:107523.

Financial Modeling Prep (2026). Financial modeling prep: Financial data & stock api. https://site.financialmodelingprep.com/developer/docs/dashboard. Data vendor API. Accessed July 2026.

French, K. R. (2026). Kenneth R. French Data Library. https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html. Accessed: 2026-07-13.

Gavriilidis, K. (2021). Measuring climate policy uncertainty. Working paper, SSRN.

Gavriilidis, K., Känzig, D. R., Raghavan, R., and Stock, J. H. (2026). The macroeconomic effects of climate policy uncertainty. NBER Working Paper 34762, National Bureau of Economic Research.

Gentzkow, M., Kelly, B., and Taddy, M. (2019). Text as data. *Journal of Economic Literature*, 57(3):535–574.

Hassan, T. A., Hollander, S., van Lent, L., and Tahoun, A. (2019). Firm-level political risk: Measurement and effects. *Quarterly Journal of Economics*, 134(4):2135–2202.

HF Data Library (2026). Hf data library: Historical financial data api. https://hfdatalibrary.com/pages/api. Data vendor API. Accessed July 2026.

Honnibal, M. and Montani, I. (2017). spaCy 2: Natural language understanding with bloom embeddings, convolutional neural networks and incremental parsing. Computer software, Explosion AI.

International Monetary Fund (2020). Global financial stability report: Markets in the time of COVID-19. Technical report, International Monetary Fund.

IPCC (2021). *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press.

IPCC (2022). *Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, Cambridge, UK and New York, NY, USA.

Jung, H. and Hannaoui, O. (2024). What do climate risk indices measure? Federal Reserve Bank of New York, Liberty Street Economics.

Kalamara, E., Turrell, A., Redl, C., Kapetanios, G., and Kapadia, S. (2020). Making text count: Economic forecasting using newspaper text. Staff Working Paper 865, Bank of England.

Krueger, P., Sautner, Z., and Starks, L. T. (2020). The importance of climate risks for institutional investors. *The Review of Financial Studies*, 33(3):1067–1111.

Landis, J. R. and Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 33(1):159–174.

Lorans, T. and Moussavi, J. (2021). Anticipating the climate change risks for sovereign bonds – part 2: Insights on the financial impacts. Research paper, FTSE Russell / LSEG.

Loughran, T. and McDonald, B. (2011). When is a liability not a liability? textual analysis, dictionaries, and 10-Ks. *Journal of Finance*, 66(1):35–65.

Munafò, M. R., Nosek, B. A., Bishop, D. V. M., Button, K. S., Chambers, C. D., Percie du Sert, N., Simonsohn, U., Wagenmakers, E.-J., Ware, J. J., and Ioannidis, J. P. A. (2017). A manifesto for reproducible science. *Nature Human Behaviour*, 1(1):0021.

National Centers for Environmental Information (2010). November 2010 national snow and ice

report. NOAA. Accessed 15 July 2026.

Ou-Yang, L. (2014). Newspaper3k: Article scraping and curation. Computer software, https://github.com/codelucas/newspaper.

Painter, M. (2020). An inconvenient cost: The effects of climate change on municipal bonds. *Journal of Financial Economics*, 135(2):468–482.

Peress, J. (2014). The media and the diffusion of information in financial markets: Evidence from newspaper strikes. *Journal of Finance*, 69(5):2007–2043.

Polanyi, L. and Zaenen, A. (2006). Contextual valence shifters. In Shanahan, J. G., Qu, Y., and Wiebe, J., editors, *Computing Attitude and Affect in Text: Theory and Applications*, pages 1–10. Springer.

Reimers, N. and Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 3982–3992. Association for Computational Linguistics.

RiskMetrics Group (1996). Riskmetrics technical document. Technical report, J.P. Morgan/Reuters.

Roberts, H., Bhargava, R., Valiukas, L., Jen, D., Malik, M. M., Bishop, C., Ndulue, E., Dave, A., Clark, J., Etling, B., Faris, R., Shah, A., Vincent, J., Sands, C., Donovan, J., and Crittenden, C. (2021). Media cloud: Massive open source collection of global news on the open web. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 15, pages 1034–1045.

Sautner, Z., van Lent, L., Vilkov, G., and Zhang, R. (2023). Firm-level climate change exposure. *Journal of Finance*, 78(3):1449–1498.

Task Force on Climate-related Financial Disclosures (2017). Final report: Recommendations of the task force on climate-related financial disclosures. Technical report, Financial Stability Board.

Zenios, S. A. (2023). The doom loop of sovereign climate risk. *Journal of Risk and Financial Management*, 16(3):176.

Appendix

This appendix is a complete evidence register for the thesis. A – Construction documents the index construction lexicons. B – Validation reports time-series diagnostics, article-level validation, and predeterminedness tests. C – Sovereign Bond Application reports specification robustness, lag-structure and dynamics checks, sub-sample stability, and multiple-testing context for the null result. D – Equity Application reports physical exposure scores, industry-level CSP betas, cross-sectional validation, SSI regime interactions, firm-level regime coefficients, and pricing tests. Programming documents the submitted code base and AI Acknowledgements states the scope of generative AI use.

A – Construction

A.1 Physical Climate Risk Lexicon

The hazard vocabulary used in scoring, organised by physical-risk category. Terms are matched as whole words; Named Storm Types intentionally overlap with Storms to capture label-based event references.

Table A.1 Physical Climate Risk Lexicon by Category

Category	**Terms**
Flooding	flood, floods, flooding, flash, fluvial, pluvial, riverine, surface, urban, coastal, lake, river, rivers
Storms	storm, storms, cyclone, cyclones, tropical, extratropical, convective, medicane, medicanes, hurricane, hurricanes, typhoon, typhoons, derecho, derechos, thunderstorm, thunderstorms, rainstorm, rainstorms, windstorm, windstorms, blizzard, blizzards, snowstorm, snowstorms, cloudburst, cloudbursts
Heat	heat, heatwave, heatwaves, dome, domes, bulb, spell, spells
Cold & Ice	cold, freeze, freezing, frost, ice, polar, vortex, snow, snowfall, winter, hard, deep
Fire	fire, fires, wildfire, wildfires, bushfire, bushfires, forest, grassland, smoke
Wind	wind, winds, gale, gales, sting, jets, jet,
Precipitation	rain, rainfall, precipitation, atmospheric, wave, waves, wet
Mass Movement	landslide, landslides, mudslide, mudslides, avalanche, avalanches, debris, flow, flows, slope, erosion, mass, movement, failure, failures
Coastal & Marine	surge, surges, coastal, post
Glacial	glacial, outburst, glof, glofs
Named Storm Types	tornado, tornadoes, tornados, typhoon, typhoons, hurricane, hurricanes, medicane, medicanes

A.2 Valence-Shifter, Intensity-Modifier, and Transition-Risk Exclusion Lexicons

This table consolidates the three auxiliary lexicons. Panels A–B govern article-level scoring; Panel C governs corpus filtering.

Table A.2 Auxiliary Lexicons: Valence Shifters, Intensity Modifiers, and Exclusions

Category	**N**	**Sources / Representative Terms**
Panel A: Valence shifters (3-token window; negator $v = -1$, amplifier $v = 1.5$, de-amplifier $v = 0.5$)		
Negators	40	Polanyi & Zaenen (2006) §3.1; Loughran & McDonald (2011)
Amplifiers	33	Polanyi & Zaenen (2006) §3.2
Deamplifiers	17	Polanyi & Zaenen (2006) §3.2
Panel B: Intensity modifiers (μ_j, 2-token window; article modifier $m_i = 1 + \sum \mu_j$)		
Catastrophic ($\mu = +1.0/+0.9$)	8	devastating, catastrophic, unprecedented, record-breaking, historic, extraordinary, apocalyptic, biblical
Severe ($\mu = +0.8/+0.7$)	14	severe, extreme, violent, powerful, deadly, fatal, destructive, ferocious, raging, torrential, punishing, relentless, crippling, paralyzing
Significant ($\mu = +0.5$)	10	major, significant, serious, widespread, massive, enormous, huge, large-scale, intense, fierce
Substantial ($\mu = +0.4$)	5	heavy, substantial, considerable, extensive, strong
Moderate ($\mu = +0.2$)	4	moderate, notable, marked, appreciable
Diminishing ($\mu = -0.3$ to -0.6)	9	minor, small, limited, localised, brief, isolated, short-lived, marginal, negligible
Panel C: Transition-risk exclusion (exact-phrase match, applied before scoring)		
Exclusion terms	13	climate policy, carbon tax, emissions trading, net zero, green transition, decarbonisation, decarbonization, transition risk, energy transition, renewable subsidy, esg investing, sustainable investing, carbon disclosure
US geographic focus	6	united states, united states of america, u.s., u.s.a., usa, america

Notes: US geographic filter retains articles with at least one focal-geography match. Exclusion terms applied as exact phrase matches before any scoring takes place.

A.3 Loughran–McDonald Negative-Sentiment Lexicon (Placebo)

Used exclusively in the placebo validity test. The article-level PCRA score is replaced by a length-normalised count of Loughran–McDonald negative terms; all downstream pipeline steps are held fixed.

Table A.3 Loughran–McDonald Lexicon Categories Used in Placebo

Category	**N**	**Representative Terms**
Negative	2,355	abandon, abnormal, adverse, breach, default, decline, impairment, loss, litigation, failure
Uncertainty	297	abeyance, ambiguity, anticipate, contingency, doubt, possible, uncertain, approximate

Notes: Full dictionary: https://sraf.nd.edu/loughranmcdonald-master-dictionary/.

B — Validation

B.1 Time-Series Properties of PCRA Components

Table B.1a Distributional and Time-Series Properties of PCRA Components

Statistic	S_t	SSI	CSP	Target / note
Panel A: Distribution				
Mean	0.00366	0.00319	−0.0028	CSP near zero ✓
Std	0.00883	0.00164	1.024	CSP near unity ✓
Skewness	+10.92	−0.01	+6.16	
Excess kurtosis	+212.8	−0.44	+59.4	Event-driven; expected
Panel B: Stationarity				
ADF p	—	0.001	< 0.001	CSP stationary ✓
KPSS p	—	0.010	0.100	Fail to reject ✓
CUSUM max stat (SSI)	—	20.10	—	Level shift 2014+
*Panel C: Orthogonality (*CSP *vs lagged* SSI*)*				
Pearson corr.	—	—	−0.059	$\lvert\rho\rvert < 0.10$ ✓
Distance corr. dcor	—	—	0.188	> 0.15 threshold; qualifies CSP
*Panel D: Serial correlation (*CSP *)*				
$\phi(1)$	—	—	+0.037	
LB $Q(1)$	—	—	6.18 ($p = 0.013$)	Marginal; resolved by AR(1)
LB $Q(5)$	—	—	10.61 ($p = 0.060$)	Pass ✓
LB $Q(10)$	—	—	15.55 ($p = 0.113$)	Pass ✓
Spectral power $<$ weekly	—	—	42.6%	
*Panel E: Conditional heteroscedasticity (*CSP *)*				
ARCH-LM(5)	—	—	0.61 ($p = 0.988$)	Pass ✓
ARCH-LM(10)	—	—	5.74 ($p = 0.837$)	Pass ✓
*Panel F: Variance ratio (*CSP *) — VR*< 1 *confirms mean-reversion*				
VR(2)	—	—	0.522 ($z = -31.84,\ p < 0.001$)	
VR(5)	—	—	0.213 ($z = -23.95,\ p < 0.001$)	
VR(10)	—	—	0.102 ($z = -17.73,\ p < 0.001$)	

Notes: Distance correlation = 0.188 exceeds the pre-set 0.15 threshold; CSP satisfies linear orthogonality to lagged SSI but retains nonlinear dependence, qualifying rather than overturning the decomposition. CSP serial-correlation properties are stable across AR detrending orders 0 through 2 (all pass Ljung–Box at lag 5).

Table B.1b: Sensitivity of SSI to the Half-Life Parameter

Half-life (bus. days)	λ	Mean	Std. Dev.	Corr. with Baseline
15	0.954842	0.003195	0.001888	0.960808
30[a]	0.977159	0.003185	0.001642	1.000000
45	0.984715	0.003175	0.001544	0.987955
90	0.992328	0.003142	0.001436	0.923785

[a] Baseline specification used throughout the main analysis ($\lambda = 0.977$).
Notes: The exponentially weighted moving average (EWMA) decay parameter relates to the half-life via $\lambda = 0.5^{1/h}$. Correlation is computed against the baseline SSI series ($h = 30$). All alternative half-lives retain correlation above 0.92 with the baseline, supporting the robustness claim in Section 5.1.

B.2 Human-Coder Audit

Precision, recall, and agreement statistics for a stratified 600-article sample (2014–2025 corpus).

Coder intensity scale. Each article was rated on a 1–5 ordinal scale: (1) physical climate risk mentioned only in passing; (2) minor but noted theme; (3) moderate theme, balanced with other topics; (4) major theme, substantially covering disaster impacts or risk exposure; (5) dominant or sole focus. Scores 4–5 are treated as coder-relevant under the physical-only rule. Country attribution used a 1–3 scale: (1) no US attribution; (2) partial US attribution; (3) predominant or exclusive US attribution.

Table B.2 Human-Coder Audit: Classification Performance

	Physical-only rule (primary)		*Combination rule*	
Metric	Value	Threshold	Value	Threshold
Confusion matrix				
True positives	308	—	302	—
True negatives	164	—	184	—
False positives	36	—	98	—
False negatives	92	—	16	—
Performance metrics				
Precision	0.895	≥ 0.85 ✓	0.755	≥ 0.85 ×
Recall	0.770	≥ 0.75 ✓	0.950	≥ 0.75 ✓
F1	0.828	—	0.841	—
Accuracy	0.787	—	0.810	—
Cohen's κ	0.551	≥ 0.60 ×	0.612	≥ 0.60 ✓
Spearman ρ (intensity)	0.194	≥ 0.60 ×	0.281	≥ 0.60 ×

Notes: Physical-only rule: coder relevance score of 1 AND country attribution = 3. Combination rule: relevance of 1 AND country ≥ 2.

B.3 Predeterminedness Tests

Three complementary tests confirm that past yield changes do not predict future CSP values, supporting a causal interpretation of the local projection estimates.

Table B.3 Predeterminedness Tests: CSP and Sovereign Yield Changes

Panel A: Granger non-causation ($|\Delta y| \not\to$ CSP), $N = 4{,}096$

Lag	F	p	Predet.
1	0.617	0.432	Yes
2	0.367	0.693	Yes
3	0.384	0.765	Yes
4	2.286	0.058	Yes
5	3.064	0.009	No[a]

Panel B: Joint Wald test lags 1, 2, 5 (HAC), $N = 4{,}096$

	Coef. (s.e.)	p
$\hat{\beta}_1$	0.015 (0.054)	0.787
$\hat{\beta}_2$	−0.056 (0.051)	0.281
$\hat{\beta}_5$	−0.074 (0.054)	0.170
Joint $F(3, 4092)$	1.063	0.364

Panel C: Distance correlation, $N = 4{,}101$

	dcor	p	Predet.
CSP, Δy	0.027	—	Yes
CSP, $\lvert\Delta y\rvert$	0.030	0.343[b]	Yes

[a] Isolated rejection; does not survive the joint Wald test or HAC standard errors ($p = 0.170$ individually under HAC).

[b] Permutation p-value ($B = 10{,}000$). SSI–yield-level distance correlation (0.576) reflects a shared low-frequency trend.

C — Sovereign Bond Application

C.1 Specification Robustness: Index Choice, Individual Controls, and Geopolitical Risk

The contemporaneous CSP coefficient on the 30Y−2Y spread ($h = 0$) under alternative index choices, individual controls added one at a time, and a geopolitical-risk (GPR) control.

Table C.1 Specification Robustness: 30Y−2Y Spread, $h = 0$

Panel A: Index choice

Spec	$\hat{\beta}_0$	p
CSP only, no ctrl	7.034×10^{-6}	0.443
CSP only, full ctrl	7.017×10^{-6}	0.445
CSP+SSI, no ctrl	6.944×10^{-6}	0.450
CSP+SSI, full ctrl	6.926×10^{-6}	0.451

Panel B: Individual controls

Spec	$\hat{\beta}_0$	p
CSP only	7.034×10^{-6}	0.443
+MOVE	7.033×10^{-6}	0.443
+CPU	7.015×10^{-6}	0.445
+EPU	7.036×10^{-6}	0.443
+all	7.017×10^{-6}	0.445

Panel C: Geopolitical risk

Spec	$\hat{\beta}_0$	p
CSP only, no ctrl	7.034×10^{-6}	0.443
+MOVE/CPU/EPU	7.017×10^{-6}	0.445
+GPR only	7.030×10^{-6}	0.443
+all incl. GPR	6.989×10^{-6}	0.446

Notes: $n = 3{,}916$ throughout. Newey–West HAC standard errors. Panel B: maximum drift from any single control is 0.3%. Panel C: nested F-test for GPR's marginal contribution over MOVE/CPU/EPU: $F = 0.274$, $p = 0.601$.

C.2 Lag-Structure and Dynamics Robustness

Table C.2 Lag-Structure and Dynamics Robustness: 30Y−2Y Spread

Panel A: CSP *lags (days)*

Lag	$\hat{\beta}$ (full ctrl)	p
1 day	-1.061×10^{-5}	0.274
3 days	-9.440×10^{-6}	0.365
5 days	-1.980×10^{-5}	0.108

Panel B: SSI *lags (weeks)*

Lag	$\hat{\beta}$	p	n
1	-1.73×10^{-3}	0.855	783
4	5.02×10^{-3}	0.526	780
8	-5.37×10^{-3}	0.523	776
12	1.85×10^{-3}	0.825	772

Panel C: AR(2) dynamics check

Metric	No lags	AR(2)
Mean $\lvert\Delta\beta_h\rvert$	9.07×10^{-6}	9.14×10^{-6}
Mean SE	10.0×10^{-6}	7.77×10^{-6}
Sig. horiz. (of 21)	1	1

Notes: Panel A/C: $n = 3{,}916$; Panel A shown for the full-control specification (no-control results are within 0.3% of the values shown throughout). No lag in Panel A or B reaches significance at the 10% level; the minimum p-value in Panel A is 0.108 at five days. Panel C: IRF volatility is unchanged after adding two lags of the dependent variable, indicating the horizon-to-horizon jumpiness in Figure 9 reflects sampling noise around a null rather than omitted dynamics.

C.3 Sub-Sample Stability

Baseline local projection, 10-year yield, full controls (MOVE, CPU, EPU).

Table C.3 Sub-Sample Stability: CSP → 10-Year Yield

Period	n	Sig. horizons (of 21)	$\max \lvert\hat{\beta}\rvert$	Direction
2008–2015	1,836	2	3.07×10^{-5}	Mostly positive
2016–2019	1,042	1	3.75×10^{-5}	Mostly positive
2020–2025	1,296	0	3.55×10^{-5}	Mixed

Notes: Newey–West HAC standard errors. The number of significant horizons declines monotonically across sub-periods and reaches zero in 2020–2025.

C.4 Multiple-Testing Context and Economic Magnitude

Table C.4a Multiple-Testing Context and Economic Magnitude

Statistic	Value
Implicit tests run across grids (10Y only)	147
Expected false positives at 5% under a true null	7.4
Observed significant horizons ($p < 0.05$)	4
Binomial test (H_0: hits = chance rate)	$p = 0.939$
CSP standard deviation	1.024
30Y−2Y spread response to a 1-SD CSP shock	0.072 bp
Nested F-test: CSP-only vs. CSP+MOVE/CPU/EPU	$F = 0.003$, $p = 0.9998$

Notes: The observed count of significant horizons (4) is below the number expected by chance alone under a true null (7.4), consistent with no systematic effect. The 1-SD economic magnitude (0.072 basis points) is negligible on any reasonable materiality threshold.

Table C.4b Multiple-Testing Context, Full Term-Structure Grid

Statistic	Value
Implicit tests run across grids (all 4 maturities)	588
Expected false positives at 5% under a true null	29.4
Observed significant horizons ($p < 0.05$)	18
Binomial test (H_0: hits = chance rate)	$p = 0.992$
2Y (of 147)	0
5Y (of 147)	7
10Y (of 147)	4
30Y (of 147)	7

Notes: Extending the count to all four maturities remains consistent with pure chance ($p = 0.992$).

C.5 Event-Study Significance Tests

Table C.5 Event Study: Day-Level, Pre/Post, and Cumulative Tests

Test	Statistic	p	n (events)
Day -10 (one-sample t-test)	$t = -1.86$	0.096	10
Pre (-10 to -1) vs. post ($+1$ to $+10$)	$t = -0.18$	0.863	10
Cumulative 5-day post-spike response	$t = -0.27$	0.797	10

Notes: One-sample and paired-comparison t-tests on the normalised 10-year yield , using event-level means to avoid pseudo-replication across overlapping windows. Of 21 relative days tested individually, only day -10 reaches significance at the 10% level.

C.6 Asymmetry Test; Full Maturity Grid

Table C.6 Asymmetry Test, All Maturities

Series	β_{pos} (p)	β_{neg} (p)	Wald p	n
30Y−2Y spread	9.03×10^{-6} (0.388)	-7.99×10^{-6} (0.838)	0.693	3,916
2Y yield	4.94×10^{-6} (0.596)	-5.87×10^{-5} (0.177)	0.179	4,176
5Y yield	5.20×10^{-6} (0.586)	-5.22×10^{-5} (0.258)	0.247	4,176
10Y yield	1.10×10^{-5} (0.288)	-7.22×10^{-5} (0.115)	0.095	4,175
30Y yield	1.13×10^{-5} (0.288)	-6.58×10^{-5} (0.110)	0.092	3,916

Notes: Wald H_0: $\beta_{\text{pos}} = \beta_{\text{neg}}$. Only the 10Y and 30Y rows approach the 10% threshold; neither the difference nor its statistical borderline status extends to the spread specification most directly motivated by the maturity-gradient hypothesis.

D — Equity Application

D.1 IPCC AR6 Physical Exposure Scores

Table D.1 IPCC Physical Exposure Scores: FF49 and FF12

ξ	Count	FF49 Industries
5	4	Agriculture, Real Estate, Utilities, Oil
4	9	Chemicals, Insurance, Paper, Coal, Mines, Transport, Food & Beverages, Construction, Shipping
3	11	Clothing, Soda, Meals, Banks, Rubber, Beer, Gold, Boxes, Textiles, Building Materials, Steel
2	15	Electrical Equipment, Finance, Chips, Wholesale, Retail, Telecom, Entertainment, Autos, Household, Fabricated Products, Tobacco, Machinery, Health, Aerospace, Other
1	10	Drugs, Lab Equipment, Software, Hardware, Business Services, Personal Services, Books, Guns, Medical Equipment, Toys
FF12 headline scores		
5		Energy, Utilities
3		Non-Durables, Manufacturing, Chemicals
2		Durables, Retail, Other
1		Business Equipment, Telecom, Health, Finance

Notes: Scores assigned following IPCC AR6 WGII (2022) qualitative vulnerability language: high-confidence high-damage ($\xi = 5$) covers agriculture, energy systems, and physical infrastructure; finance and business equipment ($\xi = 1$) face the lowest direct physical exposure. Scores are time-invariant and set before any financial estimation.

D.2 Cross-Sectional Validation: Industry CSP Betas on Physical Exposure

Table D.2 Beta Sort: Industry CSP Beta on Physical Exposure ($n = 49$ FF49 industries)

Spec	Controls	$\hat{b}$	p	R^2	Spearman ρ	p_ρ
M1	None	-2.55×10^{-5}	0.248	0.027	-0.199	0.17
M2	+SSI	-2.62×10^{-5}	0.231	0.029	-0.195	0.179
M3	+CPU	-2.61×10^{-5}	0.235	0.031	-0.159	0.179
M4	+FF5	-2.37×10^{-5}	0.260	0.023	-0.157	0.282
M5	+SSI+FF5	-2.37×10^{-5}	0.260	0.024	-0.157	0.282

Notes: HC3 standard errors. Slope is negative and Spearman rank correlation correctly signed in all five specifications including the most demanding FF5 control.

D.3 FF49 Industry CSP Betas

Full cross-section ranked from most negative to most positive within each beta tercile. M5 includes SSI, and FF5 controls.

Table D.3 FF49 Industry CSP Betas and Physical Exposure Scores

Low-beta tercile

Ind.	$\hat{\beta}$	SE	t	p	ξ
Gold	-4.06×10^{-4}	2.87×10^{-4}	-1.41	0.158	3
Soda	-3.44×10^{-4}	1.36×10^{-4}	-2.52	0.012	3
Agric	-3.16×10^{-4}	1.80×10^{-4}	-1.75	0.080	5
FabPr	-2.70×10^{-4}	2.63×10^{-4}	-1.03	0.304	2
Beer	-2.60×10^{-4}	9.50×10^{-5}	-2.75	0.006	3
Smoke	-2.60×10^{-4}	1.30×10^{-4}	-2.01	0.044	2
Coal	-2.45×10^{-4}	3.94×10^{-4}	-0.62	0.535	4
Fun	-2.05×10^{-4}	1.83×10^{-4}	-1.12	0.263	2
Food	-1.67×10^{-4}	9.60×10^{-5}	-1.74	0.082	4
Autos	-1.64×10^{-4}	2.05×10^{-4}	-0.80	0.423	2
PerSv	-1.40×10^{-4}	1.60×10^{-4}	-0.88	0.381	1
Clths	-1.27×10^{-4}	1.27×10^{-4}	-1.00	0.316	3
Whlsl	-1.27×10^{-4}	6.50×10^{-5}	-1.95	0.051	2
Chems	-1.10×10^{-4}	1.01×10^{-4}	-1.09	0.276	4
Banks	-1.03×10^{-4}	8.50×10^{-5}	-1.21	0.225	3
Fin	-1.02×10^{-4}	8.20×10^{-5}	-1.25	0.212	2
MedEq	-9.00×10^{-5}	1.08×10^{-4}	-0.83	0.406	1

Middle-beta tercile

Ind.	$\hat{\beta}$	SE	t	p	ξ
Insur	-7.80×10^{-5}	8.50×10^{-5}	-0.92	0.360	4
Paper	-7.40×10^{-5}	8.50×10^{-5}	-0.87	0.385	4
Meals	-7.00×10^{-5}	9.90×10^{-5}	-0.70	0.481	3
RlEst	-6.80×10^{-5}	1.19×10^{-4}	-0.57	0.569	5
Ships	-5.00×10^{-5}	1.26×10^{-4}	-0.40	0.690	4
Mach	-3.30×10^{-5}	8.70×10^{-5}	-0.38	0.706	2
Hlth	-2.40×10^{-5}	1.34×10^{-4}	-0.18	0.857	2
Rtail	-1.80×10^{-5}	7.50×10^{-5}	-0.23	0.814	2
Guns	-1.20×10^{-5}	1.45×10^{-4}	-0.08	0.933	1
Telcm	-1.10×10^{-5}	1.04×10^{-4}	-0.11	0.916	2
Other	-7.00×10^{-6}	7.60×10^{-5}	-0.09	0.928	2
Hardw	-6.00×10^{-6}	1.05×10^{-4}	-0.06	0.954	1
Rubbr	$+1.00 \times 10^{-6}$	1.24×10^{-4}	$+0.01$	0.990	3
Chips	$+2.00 \times 10^{-5}$	1.02×10^{-4}	$+0.19$	0.847	2
Oil	$+2.00 \times 10^{-5}$	1.55×10^{-4}	$+0.13$	0.896	5
Softw	$+3.80 \times 10^{-5}$	6.70×10^{-5}	$+0.56$	0.576	1

High-beta tercile

Ind.	$\hat{\beta}$	SE	t	p	ξ
Trans	$+3.90 \times 10^{-5}$	8.60×10^{-5}	$+0.45$	0.651	4
Util	$+4.60 \times 10^{-5}$	1.07×10^{-4}	$+0.43$	0.666	5
Txtls	$+6.30 \times 10^{-5}$	1.80×10^{-4}	$+0.35$	0.725	3
Aero	$+6.60 \times 10^{-5}$	1.36×10^{-4}	$+0.48$	0.631	2
Drugs	$+7.40 \times 10^{-5}$	1.03×10^{-4}	$+0.72$	0.470	1
Steel	$+8.90 \times 10^{-5}$	1.64×10^{-4}	$+0.54$	0.587	3
Boxes	$+9.60 \times 10^{-5}$	1.14×10^{-4}	$+0.84$	0.401	3
LabEq	$+1.01 \times 10^{-4}$	8.10×10^{-5}	$+1.25$	0.210	1
Hshld	$+1.04 \times 10^{-4}$	8.00×10^{-5}	$+1.29$	0.196	2
BldMt	$+1.06 \times 10^{-4}$	1.07×10^{-4}	$+0.99$	0.320	3
BusSv	$+1.35 \times 10^{-4}$	0.66×10^{-5}	$+2.03$	0.042	1
Books	$+1.12 \times 10^{-4}$	1.36×10^{-4}	$+0.82$	0.410	1
ElcEq	$+1.74 \times 10^{-4}$	1.41×10^{-4}	$+1.24$	0.217	2
Toys	$+2.06 \times 10^{-4}$	1.53×10^{-4}	$+1.34$	0.180	1
Cnstr	$+2.66 \times 10^{-4}$	1.67×10^{-4}	$+1.59$	0.111	4
Mines	$+2.39 \times 10^{-4}$	1.68×10^{-4}	$+1.42$	0.154	4

Notes: $N \approx 4{,}200$ per industry; NW-HAC standard errors. M5 controls: SSI & FF5. Industries ordered from most negative to most positive $\hat{\beta}$ within each tercile.

D.4 SSI Regime Interaction Coefficients ($\hat{\gamma}_i$)

Full FF49 cross-section. Per-industry incremental CSP response during high-SSI periods ($SSI_{t-1} > p75$), estimated by per-industry OLS with HAC standard errors. Industries ranked most negative to most positive $\hat{\gamma}_i$ within each exposure tier.

Table D.4 SSI Regime Interaction Coefficients $\hat{\gamma}_i$: Full FF49 Cross-Section

High exposure

Ind.	ξ	$\hat{\gamma}_i$	p
Food	4	−5.65	0.037 **
Insur	4	−5.04	0.146
RlEst	5	−4.47	0.145
Util	5	−4.45	0.011 **
Oil	5	−4.21	0.200
Ships	4	−4.12	0.464
Trans	4	−3.96	0.328
Paper	4	−3.16	0.378
Chems	4	−2.62	0.519
Cnstr	4	−2.54	0.628
Agric	5	+0.23	0.930
Mines	4	+5.61	0.365
Coal	4	+8.75	0.471

Medium exposure

Ind.	ξ	$\hat{\gamma}_i$	p
Banks	3	−26.02	0.151
BldMt	3	−26.01	0.138
Boxes	3	−22.99	0.143
Clths	3	−17.89	0.313
Rubbr	3	−15.83	0.267
Beer	3	−12.92	0.267
Txtls	3	−9.02	0.689
Autos	2	−6.55	0.658
Gold	3	−5.55	0.822
Smoke	2	−4.51	0.580
Meals	3	−3.59	0.761
Soda	3	−0.56	0.963
Telcm	1	+2.37	0.718
Hshld	1	+4.70	0.464
Fin	1	+7.87	0.405
ElcEq	2	+8.68	0.406
Aero	2	+9.06	0.392
Rtail	2	+9.79	0.206
Whlsl	2	+10.10	0.184
Steel	3	+11.27	0.684
Fun	2	+11.82	0.314
Mach	2	+12.35	0.240
Other	2	+12.54	0.115
Hlth	2	+13.37	0.181
Chips	2	+15.52	0.108
FabPr	2	+19.98	0.218

Low exposure

Ind.	ξ	$\hat{\gamma}_i$	p
Toys	1	+0.86	0.839
PerSv	1	+1.53	0.672
LabEq	1	+2.83	0.389
BusSv	1	+2.92	0.351
Drugs	1	+3.37	0.201
Hardw	1	+5.00	0.141
MedEq	1	+5.06	0.081 *
Softw	1	+5.80	0.088 *
Guns	1	+5.92	0.046 **
Books	1	+7.32	0.092 *

Notes: $\hat{\gamma}_i$ (in units of $\times 10^{-4}$) is the coefficient on $CSP_t \times 1[SSI_{t-1} > p75]$ with HAC standard errors.

D.5 Firm-Level SSI Regime Interaction Coefficients: Food and Utilities Constituents

Individual firm-level regime coefficients for all constituent firms in the two industries identified as individually significant. Sorted by $\hat{\gamma}_i$ ascending (most amplifying first) within each industry.

Table D.5a Food Constituents: Firm-Level CSP and Regime Betas ($n = 14$ firms)

Ticker	Firm	n	$\hat{b}_{\text{CSP}}$	p_{CSP}	$\hat{\gamma}_i$ (regime)	p_γ
OTLY	Oatly Group	1,156	0.1964	0.250	−0.2363	0.519
SJM	J.M. Smucker	4,283	0.0128	0.548	−0.1498	0.004**
LW	Lamb Weston	2,299	0.0294	0.628	−0.1396	0.080*
CPB	Campbell's	4,283	0.0339	0.296	−0.1348	0.021**
BYND	Beyond Meat	1,675	0.4893	0.213	−0.1218	0.811
TSN	Tyson Foods	4,283	0.0223	0.422	−0.1192	0.080*
KHC	Kraft Heinz	2,638	−0.0152	0.605	−0.1034	0.083*
GIS	General Mills	4,283	0.0137	0.452	−0.0810	0.047**
MDLZ	Mondelez Int'l	4,284	0.0155	0.454	−0.0735	0.044**
ADM	Archer-Daniels-Midland	4,283	−0.0017	0.957	−0.0705	0.259
HRL	Hormel Foods	4,283	−0.0100	0.622	−0.0625	0.200
CAG	Conagra Brands	4,283	0.0019	0.932	−0.0510	0.357
HSY	Hershey	4,283	0.0086	0.686	−0.0254	0.545
MKC	McCormick & Co.	4,283	−0.0048	0.810	−0.0044	0.920

Notes: 14/14 firms negative; 7 significant at $p < 0.10$ (marked *), 4 at $p < 0.05$ (marked **). Mean firm-level $\hat{\gamma}_i = -0.0981$, versus pooled firm-FE panel estimate $\hat{\gamma} = -0.0842$ (SE 0.0284, $p = 0.003$, $n = 50{,}599$ firm-days).

Table D.5b Utilities Constituents: Firm-Level CSP and Regime Betas ($n = 34$ firms)

Ticker	Firm	n	$\hat{b}_{\text{CSP}}$	p_{CSP}	$\hat{\gamma}_i$ (regime)	p_γ
WMB	Williams Cos.	4,283	0.1321	0.166	−0.2231	0.048**
ET	Energy Transfer	4,283	0.0981	0.168	−0.1816	0.086*
NRG	NRG Energy	4,283	0.0303	0.423	−0.1610	0.103
CEG	Constellation Energy	980	−0.1195	0.209	−0.1456	0.624
PPL	PPL Corp.	4,283	0.0244	0.163	−0.1246	0.006**
PNW	Pinnacle West	4,283	0.0344	0.049**	−0.1219	0.007**
ED	Consolidated Edison	4,283	0.0221	0.194	−0.1151	0.004**
CLNE	Clean Energy Fuels	4,284	0.0505	0.529	−0.1151	0.444
PEG	Public Serv. Enterprise	4,283	0.0488	0.020**	−0.1093	0.010**
CMS	CMS Energy	4,283	0.0368	0.037**	−0.1029	0.006**
ETR	Entergy	4,283	0.0331	0.105	−0.1017	0.024**
AEP	American Electric Power	4,284	0.0423	0.045**	−0.1012	0.006**
AEE	Ameren	4,283	0.0380	0.069*	−0.0980	0.020**
FE	FirstEnergy	4,283	0.0412	0.060*	−0.0969	0.017**
NI	NiSource	4,283	0.0331	0.121	−0.0962	0.036**
CNP	CenterPoint Energy	4,283	0.0257	0.255	−0.0932	0.086*
XEL	Xcel Energy	4,284	0.0410	0.029**	−0.0918	0.016**
DUK	Duke Energy	4,283	0.0267	0.078*	−0.0909	0.009**
EXC	Exelon	4,284	0.0319	0.141	−0.0888	0.056*
EIX	Edison Int'l	4,283	0.0126	0.499	−0.0874	0.059*
SO	Southern Co.	4,283	0.0385	0.039**	−0.0807	0.026**
ATO	Atmos Energy	4,283	0.0222	0.180	−0.0767	0.066*
OKE	ONEOK	4,283	0.0162	0.666	−0.0745	0.331
WEC	WEC Energy Group	4,283	0.0275	0.116	−0.0735	0.052*
D	Dominion Energy	4,283	0.0106	0.601	−0.0701	0.095*
DTE	DTE Energy	4,283	0.0270	0.124	−0.0696	0.043**
LNT	Alliant Energy	4,284	0.0220	0.237	−0.0685	0.088*
SRE	Sempra	4,283	0.0304	0.156	−0.0683	0.098*
ES	Eversource Energy	4,283	0.0122	0.528	−0.0659	0.090*
KMI	Kinder Morgan	3,741	0.0127	0.751	−0.0607	0.379
EVRG	Evergy	4,283	0.0199	0.314	−0.0501	0.194
NEE	NextEra Energy	4,283	0.0249	0.288	−0.0375	0.305
AWK	American Water Works	4,282	0.0113	0.575	−0.0368	0.350
PCG	PG&E Corp.	4,283	0.0483	0.097*	−0.0308	0.783

Notes: 34/34 firms negative; 25 significant at $p < 0.10$ (marked *), 15 at $p < 0.05$ (marked **). Mean firm-level $\hat{\gamma}_i = -0.0944$, versus pooled firm-FE panel estimate $\hat{\gamma} = -0.0922$ (SE 0.0331, $p = 0.005$, $n = 141{,}781$ firm-days). All specifications use firm fixed effects, standard errors two-way clustered by firm and date.

D.6 Cross-Sectional Pricing Tests: Quintile Sorts on CSP Sensitivity

Zero-cost spread portfolios (Q5−Q1) formed by sorting the 138 constituent firms from the six industries examined at firm level into quintiles on trailing 63-day rolling CSP betas, re-estimated monthly. The unconditional ranking sorts on the raw CSP beta; the conditional ranking sorts on the effective high-SSI slope ($\hat{b}_{\text{CSP}} + \hat{b}_{\text{interaction}}$).

Table D.6a Monthly Spread-Portfolio Alphas: Unconditional vs. High-SSI-Conditional Ranking

	Unconditional ranking		Conditional (high-SSI) ranking	
Factor model	α (%/mo)	p	α (%/mo)	p
Market	+0.31	0.646	−0.22	0.692
FF3	+0.32	0.621	−0.16	0.767
FFC (Carhart)	+0.28	0.640	−0.08	0.878
FF5	+0.21	0.709	−0.14	0.803
FF5+UMD	+0.19	0.727	−0.07	0.904

Notes: $n = 92$ months. Conditional ranking uses the effective slope from a rolling regression of $\text{ret}_{\text{excess}}$ on CSP, CSP × HighSSI, HighSSI, and Carhart controls, with HighSSI defined globally at the 75th percentile of SSI_{t-1} (= 0.00445; 25.0% of firm-days).

Table D.6b Daily and Weekly Spread-Portfolio Alphas, Same Quintile Rankings

	Unconditional ranking		Conditional (high-SSI) ranking	
Factor model	Daily α (%/day, ann.)	p	Daily α (%/day, ann.)	p
Market	−0.0055 (−1.38%)	0.702	+0.0192 (+4.95%)	0.384
FF3	−0.0060 (−1.49%)	0.678	+0.0203 (+5.25%)	0.354
FFC	−0.0063 (−1.58%)	0.659	+0.0193 (+4.98%)	0.376
FF5	−0.0052 (−1.29%)	0.716	+0.0186 (+4.79%)	0.399
FF5+UMD	−0.0053 (−1.33%)	0.707	+0.0180 (+4.64%)	0.410
	Weekly α (%/wk, ann.)		Weekly α (%/wk, ann.)	
Market	−0.0374 (−1.93%)	0.576	+0.0492 (+2.59%)	0.380
FF3	−0.0442 (−2.27%)	0.510	+0.0495 (+2.61%)	0.379
FFC	−0.0478 (−2.46%)	0.472	+0.0433 (+2.28%)	0.444
FF5	−0.0406 (−2.09%)	0.539	+0.0431 (+2.27%)	0.451
FF5+UMD	−0.0427 (−2.20%)	0.516	+0.0392 (+2.06%)	0.496

Notes: Daily: n = 4,214 (unconditional), 1,930 (conditional) trading days. Weekly: n = 875 (unconditional), 722 (conditional). Quintile assignment uses monthly-estimated rankings held fixed within each holding period (up to one month stale); this is not a re-estimation at daily/weekly frequency. No alpha is significant at any frequency, ruling out monthly-compounding as the source of the monthly-horizon null.

D.7 Sector-Restricted Spread Portfolio Alphas

The pooled sort in Table D.6 ranks all 138 firms against peers across six unrelated industries, so a firm's quintile depends in part on its CSP beta relative to firms outside its own sector. This table repeats the identical procedure, rolling 63-day CSP beta, re-estimated monthly, Newey–West standard error, restricted separately to firms within each industry, so rankings compare only same-sector peers.

Table D.7 Sector-Restricted Spread-Portfolio Alphas

Industry	Factor model	α (%/mo)	t	p	n firms	n months
Food	Market	+2.24	1.00	0.317	14	202
Food	FF3	+2.44	1.09	0.275	14	202
Food	FFC	+2.23	1.10	0.269	14	202
Food	FF5	+2.04	1.09	0.276	14	202
Food	FF5+UMD	+1.87	1.09	0.274	14	202
Util	Market	+0.12	0.42	0.677	34	202
Util	FF3	+0.09	0.32	0.748	34	202
Util	FFC	+0.11	0.39	0.694	34	202
Util	FF5	+0.05	0.18	0.857	34	202
Util	FF5+UMD	+0.07	0.26	0.799	34	202

Notes: No alpha is statistically distinguishable from zero in either sector under any factor model. Food shows the largest point estimate of the tradability analysis overall, but its standard error is wide enough to leave it indistinguishable from zero.

Programming

The zip file `PCRA Index Thesis - Adam Mc Guire (770113).zip` contains all programming files used to construct and validate the PCRA index and to perform the subsequent analyses. Each notebook is independent, loading its inputs directly from disk rather than from a shared kernel, so that any notebook can be re-run on its own once the index exists. The sole hard dependency is that `Index_Construction.ipynb` must have been run at least once to produce the index files consumed by every other notebook. This section describes each file in order of execution.

`Index_Construction.ipynb` is the production pipeline that builds the PCRA index for the United States over the full 2008–2025 sample, comprising article retrieval (Stage A), syndication cleanup (Stage B), article-level scoring (Phase 2), novelty weighting (Phase 3), and the full sample production run.

`Article_Level.ipynb` performs two tasks: an incremental reconstruction of the index at the article level, covering volume count, uniformity, novelty, and EM-DAT coincidence, and the human-audit workflow used to validate the article-classification step, comprising filter-pass and filter-reject sampling, direct and combination scoring, and Spearman range checks.

`Index-Level_Validity.ipynb` constitutes the core statistical validation chapter, covering SSI construction and CSP distributional checks, the orthogonality check, stationarity and whiteness diagnostics, ARCH-LM, CUSUM, spectral density, half-life and AR-order sensitivity, window/offset sensitivity, the EM-DAT permutation test, and convergent validity, together with the numbered figures reported in the thesis.

`Discriminant_Validity.ipynb` tests whether CSP is genuinely distinct from CPU and EPU, through pairwise and full correlation matrices, lead-lag cross-correlations, surprise overlap, rolling correlation, distributional comparison, and PCA across CSP, CPU, and EPU.

`Predeterminedness.ipynb` checks whether CSP is predetermined with respect to same-day yield moves, using Granger causality, a joint Wald test, distance correlation, and an accompanying visual and summary check.

`Placebo_Test.ipynb` constructs a placebo SSI/CSP series using the Loughran–McDonald negative-sentiment lexicon in place of the physical-risk lexicon, applies the same EM-DAT coincidence permutation test and PCA, and compares the placebo against the true index as a falsification check.

`Sovereign_Investigation.ipynb` implements the bond-side asset-pricing application. It builds the master panel and estimates local projections on Treasury yields at the 2, 5, 10, and 30-year maturities, covering the baseline and long-horizon specifications, the term-structure gradient, impulse-response plots, response-surface heatmaps, the yield-spread local projection, sub-sample stability, SUR joint tests, a CSP-versus-SSI pricing comparison, event studies around the largest CSP spikes, asymmetry tests, and lag-structure robustness, including a geopolitical-risk cross-check.

`Industry_Investigation.ipynb` implements the equity-side asset-pricing application. It constructs FF12 and FF49 industry exposure scores and panels, estimates the daily panel regressions, compares taxonomies, and tests temporal-aggregation and cross-validation robustness, sub-sample stability, and lagged-return sensitivity, decomposing results by exposure gamma, power/severity/volatility, firm level, and salience conditionality, together with a Faccini-style monthly/weekly comparison.

AI Acknowledgements

Generative AI tools, including ChatGPT and Claude, were used in a limited supportive capacity throughout the work. Their use was focused on brainstorming, debugging and refining Python code and improving the formatting of LaTeX tables.

Literature recommendations by these tools were independently reviewed, verified, and fully understood before being incorporated into the work. The research decisions, selection and application of the methodology, analysis, interpretation of the findings, and the development of the final arguments in this thesis remain the author's own.